

Machine Forecast Disagreement in the Cryptocurrency Market*

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Abstract

This paper uses machine learning models to construct a crypto-level measure of simulated investor belief disagreement (MFD) for cryptocurrencies. Simulated investors use their individual models and common data information to form their individual return forecasts. Consistent with Miller (1977), we find a significantly negative relation between MFD and future cryptocurrency returns in the cross section. Past return variables are the top drivers of MFD, suggesting that disagreement is strongly associated with information contained in past returns. Moreover, the negative MFD-return relation is stronger for cryptocurrencies with larger limits-to-arbitrage or more severe overpricing, highlighting the role of mispricing and limits-to-arbitrage.

JEL Classification: G11, G12, G17, G40

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I. Introduction

Cryptocurrencies have been the most popular and important emerging assets since the rapid growth in recent years, despite ongoing debate about their intrinsic values (e.g., Böhme, Christin, Edelman and Moore, 2015; Griffin and Shams, 2020; Howell, Niessner and Yermack, 2020; Sockin and Xiong, 2023; Li, Shin, and Wang, 2025).¹ Unlike traditional financial assets such as equities and bonds, cryptocurrencies do not have well-accepted fundamental collaterals. Therefore, it is challenging to precisely price cryptocurrencies. Recent studies use some common asset factors such as size and past returns to identify the relative valuation of cryptocurrencies in the cross section (e.g., Borri and Shakhnov, 2022; Liu, Tsyvinski and Wu, 2022; Fieberg et al., 2025; Lee and Wang, 2025), and some studies use cryptocurrency-specific factors (e.g., Cong, Li and Wang, 2021; Liu and Tsyvinski, 2021; Sockin and Xiong, 2023; Babiak and Bianchi, 2025).

In this paper, we examine the relative valuation of cryptocurrencies in the cross section from the perspective of heterogeneous beliefs. Specifically, we follow the approach in Bali, Kelly, Mörke and Rahman (2026) to construct a measure of simulated investor disagreement on the future values of cryptocurrencies.² We argue that this machine forecast disagreement (MFD) measure is particularly suitable for cryptocurrencies. There

¹ According to CoinMarketCap, the aggregate market capitalization of cryptocurrencies around the world has been \$2.67 trillion until April 2025.

² In this paper, “investor” refers to hypothetical belief agents used to construct the MFD measure, rather than observed or identifiable cryptocurrency market participants. Consequently, investor disagreement should be interpreted as disagreement across simulated beliefs inferred from the model, rather than disagreement among actual investor in the cryptocurrency market.

is no widely accepted proxy for investor disagreement in the crypto market. Researchers could use price or earnings forecasts issued by professional analysts to construct proxies for disagreement for stocks, but analyst coverage for cryptocurrencies is limited (Cong et al., 2021). Therefore, we cannot construct a disagreement measure for most cryptocurrencies based on analyst forecast data as prior studies do for stocks (e.g., Diether, Malloy, and Scherbina, 2002).

Moreover, because of a lack of information about future cash flows and expected amounts of cryptocurrencies, it is harder for investors to precisely price cryptocurrencies than traditional assets. Investor disagreement arises because of a lack of fundamental information. In particular, because the cryptocurrency market is dominated by retail investors who have limited information processing ability and suffer from various behavioral biases (Kogan, Makarov, Niessner and Schoar, 2024), investor disagreement on cryptocurrencies' values would be large. Therefore, we need a novel approach to measure investor disagreement for cryptocurrencies. The machine forecast disagreement measure proposed by Bali et al. (2026) simulates the distribution of a large number of different beliefs by endowing each simulated investor with a (machine learning) model with random variation in model specification across investors.³ That is, MFD directly simulates a number of investors with different beliefs in a market. As a matter of fact, Bali et al. (2026) show that their MFD is better than other proxies such as analyst forecast disagreement in

³ We emphasize that MFD is a flexible belief-simulation framework that can accommodate various simple and sophisticated belief formation rules.

identifying the relation between belief disagreement and future stock returns in the cross section.

Following the approach in Bali et al. (2026), we construct a novel measure of belief disagreement among simulated investors in the cryptocurrency market. Our results show that MFD has a strong return predictability in the cross section for cryptocurrencies. We find that cryptos with higher MFD at the portfolio formation week tend to experience significantly lower returns in the subsequent week compared with those with lower MFD.⁴ Specifically, the decile portfolio of cryptos with the highest MFD significantly underperforms the decile portfolio of cryptos with the lowest MFD by 1.79% per week in terms of value-weighted market-adjusted returns. The significant return predictability of MFD is robust after controlling for well-established factors in bivariate portfolio analyses and Fama-MacBeth regressions. Our main results hold in various robustness tests.

Moreover, the MFD approach allows us to identify the most important factors driving the disagreement among investors. Because retail investors dominate in the crypto market and they tend to extrapolate past returns (e.g., Da, Huang, and Jin, 2021; Giglio, Maggiori, Stroebel, and Utkus, 2021), we examine whether a return-based MFD that only relies on past-return variables has a strong return predictability.⁵ By comparison, we also construct a non-return-based MFD that relies on other variables such as size, age, liquidity, etc. Our results show that the return-based MFD has a stronger return predictability than the non-

⁴ The terms cryptocurrency and crypto are used interchangeably in this paper.

⁵ We thank the anonymous referee for proposing this insightful suggestion.

return-based MFD. Moreover, a return-plus-based MFD that adds additional non-return information does not have a stronger return predictability than the return-based MFD. These results suggest that past-return information plays an important role in generating disagreement in the crypto market.

Then we explore the sources of the return predictability of MFD. The negative relation between MFD and expected crypto returns is consistent with the theoretical model of Miller (1977) that asset prices are upward biased when investors disagree with asset value and short-sale constraints exist for pessimistic investors. Consistent with the argument of Miller (1977), we find that the negative relation between MFD and future returns is significantly stronger among cryptos with higher limits-to-arbitrage. The high-low MFD portfolio has an average weekly value-weighted excess return of -0.71% (t-value is -1.29) and -3.74% (t-value is -3.05) among cryptos with low and high limits-to-arbitrage, respectively.⁶

In addition, we find that the negative MFD-return relation is more pronounced among cryptos with higher mispricing index,⁷ suggesting that mispricing explains the MFD-return relation. The high-low MFD portfolio has an average weekly value-weighted excess return of -0.95% (t-value is -1.82) and -2.46% (t-value is -3.16) among low- and high-

⁶ Because limits to arbitrage such as arbitrage risk and short-sale constraints in the absolute value are still high for cryptocurrencies in the low limits-to-arbitrage tercile portfolio in our analysis, the MFD-return relation is significantly negative among cryptocurrencies in the low limits-to-arbitrage tercile portfolio.

⁷ Following Stambaugh, Yu and Yuan (2015), we establish cryptocurrency mispricing score (MISP) on two well-documented return predictors in the cryptocurrency literature: size and momentum (Liu et al., 2022). The mispricing measure (MISP) is defined as the arithmetic average of the ranks of the two crypto return predictors, and higher (lower) MISP indicates overvaluation (undervaluation). Detailed definition please see Section IV.D.

mispriced cryptos, respectively. Correspondingly, we also find that the negative MFD-return relation is significantly weaker following low crypto sentiment periods than following high sentiment periods. Specifically, the high-low MFD portfolio has an average weekly value-weighted excess return of -1.16% (t-value is -1.43) and -2.87% (t-value is -3.44) following low and high sentiment periods, respectively.

The negative MFD-return relation remains in various robustness tests. In particular, we compare MFD with other proxies for disagreement in the cryptocurrency market. In particular, Garfinkel, Hsiao and Hu (2025) construct a turnover-based variable to measure investor belief disagreement for cryptocurrencies. They show that their turnover measure significantly and negatively predicts crypto returns in the cross section in the daily frequency. Our results show that MFD still significantly predicts future returns even after controlling for turnover, while turnover becomes weak in predicting returns in the weekly frequency.

Our study contributes to the literature in three main ways. First, to our best knowledge, we are the first to propose a novel and valid measure of belief disagreement for cryptocurrencies, which is different from traditional proxies such as abnormal turnover in the crypto market (Garfinkel et al., 2025). The return predictability of crypto MFD remains after controlling for abnormal turnover, which does not significantly predict returns in our setting. That is, we contribute to the emerging literature on the pricing of cryptocurrencies

by adding a new return predictor.⁸ Even controlling for these crypto characteristics, MFD still significantly and negatively predicts future cryptocurrency returns in the cross section. Moreover, because MFD as a flexible belief-simulation framework, we provide evidence that past-return information plays an important role in generating disagreement in the crypto market.

Second, we adopt machine learning models to develop a novel and valid measure of belief disagreement for the cryptocurrency market, supporting the usefulness of machine learning models in finance, especially research in cryptocurrencies. In particular, our study complements prior studies that use machine learning models to construct a measure of belief disagreement for stocks (Bali et al., 2026), corporate culture (Li, Mai, Shen and Yan, 2021), cyber risk (Jiang, Khanna, Yang and Zhou, 2024), firm-level climate risk exposure (e.g., Li, Shan, Tang and Yao, 2024), and cryptocurrency returns (e.g., Cakici, Shahzad, Bedowska-Sojka, and Zaremba, 2024). We differ from Cakici et al. (2024) by using machine learning models to construct a new pricing factor.

Third, our study contributes to the debate on the relation between belief disagreement and future returns. Existing theoretical and empirical studies offer two opposite views. On one hand, some studies such as Merton (1987) and Johnson (2004) argue that disagreement is a source of risk and should predict positive future returns. On the other hand, Miller

⁸ Prior studies document some significant cryptocurrency return predictors, including size, age, price, momentum, price volume, price volatility, failure risk, past returns of peer cryptos, network and production factors, and so on (e.g., Cong et al., 2021; Liu and Tsyvinski, 2021; Liu et al., 2022; Borri and Shakhnov, 2022; Lyandres, Palazzo and Rabetti, 2022; Zhu, Ma, and Tu, 2025).

(1977) and others argue that the market prices reflect more about the opinions of optimistic investors when short-sale constraints are binding. Prior empirical studies provide mixed evidence in the stock and bond markets. Our study provides novel evidence from the cryptocurrency market to support the argument on the negative relation between investor disagreement and future returns.

II. Data and Variables

We obtain cryptocurrency daily price and trading data from CoinMarketCap, the most reliable source for publicly available cryptocurrencies. It provides cryptocurrency daily data, including prices (open, close, high, and low), trading volume, and market capitalization. A key advantage of CoinMarketCap is its comprehensive coverage of both active and defunct cryptocurrencies (Liu et al., 2022), thus mitigating survivorship bias.

Our sample comprises 3024 cryptocurrencies from January 2013 to June 2024. Given that the majority of cryptocurrencies were created and listed after the final week of 2014, we begin our sample period in January 2015. Following common practices to address issues related to illiquidity and micro-cap bias (e.g., Bali et al., 2026), we exclude cryptocurrencies with daily price under \$0.1, daily trading volume less than \$100,000, market capitalization below \$1 million, and trading history shorter than 52 weeks.⁹

⁹ The reason is that newly born cryptocurrencies may harm the overall estimation accuracy of our analyses because of the great volatility.

To construct MFD, we use 40 cryptocurrency-specific characteristics as the complete information set,¹⁰ and randomly select a subset of 24 characteristics to represent an investor-specific incomplete information set. The MFD for week t is calculated based on characteristics during the week $t - 1$, using a 104-week (2-year) rolling window to estimate the random forecast regressor. This procedure yields an MFD measure over an out-of-sample period from January 2017 to June 2024. Finally, to ensure a sufficient cross-sectional sample for asset pricing tests, we restrict our empirical analysis to the period January 2019 to June 2024.

A. MFD Construction

To model the belief formation process, we adopt the framework of Gu, Kelly and Xiu (2020) to describe the relationship between cryptocurrency expected returns and characteristics. The general prediction model is as follow:

$$r_{i,t+1} = E_t[r_{i,t+1}] + \varepsilon_{i,t+1} \quad (1)$$

where $r_{i,t+1}$ denotes the return of the cryptocurrency $i = 1, \dots, I$, in week $t = 1, \dots, T$, and $E_t[r_{i,t+1}]$ is the investor-formed expectation of the return based on available information:

$$E_t[r_{i,t+1}] = g(z_{i,t}) \quad (2)$$

Here, $z_{i,t}$ represents the information set for cryptocurrency i in the time t . $g(\cdot)$ is a

¹⁰ We provide the detailed information of 40 characteristics in the Appendix A.

return-generating function that maps characteristics to return expectation.

Investor belief disagreement is commonly attributed to two main sources, information asymmetry (Grossman and Stiglitz, 1976; Kyle, 1985; Wang, 1994) and heterogeneous priors (Kandel and Pearson, 1995; Hong and Stein, 2007). To incorporate both channels, we propose a belief-generating model in which differences in return expectations arise from variation in investors' access to and interpretation of information.

To capture information asymmetry, we assume that investors observe different subsets of the complete information set. Each investor k has access to an incomplete information set:

$$z_{k,i,t} \in R^{d_k}, 1 \leq d_k < d \quad (3)$$

where $z_{k,i,t}$ is a subset of complete $d - dimension$ information set $z_{i,t}$, and the dimensionality d_k varies across investors. Each investor accesses the information set through a personalized selection of characteristics:

$$z_{k,i,t} = [z_{i,t}^{l_{k,1}}, \dots, z_{i,t}^{l_{k,d_k}}] \quad (4)$$

where $z_{i,t}^l \in R$ and denote the $l - th$ dimension of $z_{i,t}$, $l_k = \{l_{k,i} = \{1, \dots, d\}\}$, and $|l_k| = d_k$. The dimension of the investor-specific information might be different. This setup reflects heterogeneous attention, behavioral bias, and varying information processing capability across investors.

To simulate heterogeneous priors, we model each simulated investor's belief

formation using a random forest regression in line with Gu et al. (2020) and Bali et al. (2026). Random forest regressions are particularly effective for capturing non-linearities and complex interactions in high-dimensional data and well-suited to represent investor-specific forecasting processes. Importantly, random forest regression simulates heterogeneous priors across simulated investors by introducing a heterogeneous layer that randomizes regression specifications across simulated investors through the use of bootstrapping and dropout.

Each investor k forecasts future returns using a random forest model trained on their unique subset $z_{k,i,t}$:

$$g_k(z_{k,i,t}) = RF_k(z_{k,i,t}) \quad (5)$$

The random forest function is defined as:

$$RF_k(z_{k,i,t}) = \frac{1}{N} \sum_n^N h_n^L(z_{k,i,t}; D^{(n)}, \Theta^{(n)}) \quad (6)$$

where N is the total number of decision trees in the forest, $h_n^L(\cdot)$ is the prediction function of the n -th regression tree with the maximum depth L , $D^{(n)}$ is the bootstrap sample used to train tree n (randomly drawn with replacement from the original dataset, and $\Theta^{(n)}$ is the set of random feature selection decisions used to build tree n (i.e., which features are considered at each node split).

To implement this model empirically, we define the complete information set $z_{i,t}$ as a vector of 40 cryptocurrency characteristics. Each investor k is assigned an incomplete

information set of dimensions, which are randomly selected 24 characteristics from the full set of 40 characteristics. We simulate $K = 100$ investors, each using a random forest with the following hyperparameters: the maximum tree depth is 3, the number of trees in the ensemble is 2000, the friction of features is 0.1, and the friction of the sample is 0.05.

Following Kelly, Pruitt and Su (2019) and Gu et al. (2020), we rank all characteristics cross-sectionally each week and map them to the interval $[-1, +1]$. When training the random forest regressors, return outliers are winsorized at 1st and 99th percentiles cross-sectionally. Each random forest model is estimated using a 104-week (2-year) rolling window and is refitted every 12 weeks to reduce computational burden. Finally, investor belief disagreement is measured by the machine forecast disagreement (MFD), defined as the cross-sectional standard deviation of the 100 investor-specific return forecasts for each cryptocurrency in a given week.

$$MFD_{i,t} = \sqrt{\frac{1}{100} \sum_{k=1}^{100} (RF_{i,k,t} - \overline{RF_{i,t}})^2} \quad (7)$$

B. Control Variables

Following the existing literature on investor disagreement and cryptocurrencies (e.g., Liu and Tsyvinski, 2021; Liu et al., 2022; Kogan et al., 2024; Bali et al., 2026; Garfinkel et al., 2025), we incorporate a broad set of characteristics that are documented to explain the cross-section of expected cryptocurrency returns.

We begin with market beta (*Beta*), a fundamental factor in asset pricing, computed as the slope coefficient from a regression of daily cryptocurrency returns on the value-weighted market returns over the prior 90 days. To control size effects, we follow Liu et al. (2022) and use the logarithm of market capitalization (*Size*) as a control variable.

To account for momentum effect, we follow Liu et al. (2022) and include the past-week return (*MOM*) to mitigate potential confounding influences on the MFD-return relation. Motivated by Atilgan, Bali, Demirtas and Gunaydin (2020), we also include the abnormal trading volume (*Abvol*) as control variable, defined as the difference between the logarithm of dollar trading volume in a given week and the average of the logarithm of trading volumes over the preceding 12 weeks.

We further control for several liquidity and risk related variables. We follow Amihud (2002) and define illiquidity (*ILLIQ*) as the average weekly ratio of the absolute daily returns to dollar trading volume. Following Ang, Hodrick, Xing and Zhang (2009), we define idiosyncratic volatility (*Ivol*) as the standard deviation of residuals from a regression of excess cryptocurrency returns on excess market returns over the previous 90 days. Moreover, we use the standard deviation of daily returns over the same period to control for return volatility (*Volt*).

To capture investor preferences for lottery-like payoffs, we follow Bali, Cakici and Whitelaw (2011) and use the average of the five highest daily returns over the prior five weeks to control for *MAX*. We also include turnover (*Turnover*), calculated as the

logarithm of the ratio of average daily dollar volume to the market capitalization (Chordia, Subrahmanyam and Anshuman, 2001). Lastly, to account for asymmetry in return distributions, we use the coefficient on the squared excess market return term from a regression of daily excess cryptocurrency returns on both the linear and squared excess market returns over the past 90 days to control for co-skewness (*Coskew*).

C. Descriptive Statistics

Figure 1 presents the weekly time-series of aggregate MFD in the cryptocurrency market. The plot reveals that the cross-sectional median MFD tends to rise during periods of adverse market conditions, suggesting that disagreement intensifies in market downturns.

<Insert Figure 1 Here>

To capture the cross-sectional distribution for MFD and other characteristics of cryptocurrencies, we first compute weekly cross-sectional statistics and Table 1 reports weekly time-series averages. As Table 1 shows, MFD exhibits a time-series mean of 0.10, a median of 0.08, and a standard deviation of 0.07. The substantial cross-sectional dispersion in MFD is noteworthy, indicating considerable heterogeneity in investor disagreement across cryptocurrencies. This pronounced right-skewness in the MFD distribution aligns with the findings of Garfinkel et al. (2025).

<Insert Table 1 Here>

Table 2 reports the cross-sectional Spearman rank correlation coefficients between

MFD and other control variables. Consistent with theoretical predictions and empirical evidence (e.g., Bali et al., 2026), we observe a negative association between MFD and one-week-ahead excess returns, suggesting that higher disagreement is linked to lower future returns. Specifically, cryptocurrencies with smaller size, lower liquidity, stronger momentum, higher abnormal trading volume, lower market beta, higher idiosyncratic and total return volatility, greater lottery demand, higher turnover, and lower co-skewness tend to exhibit higher MFD values.

In particular, we find a strong negative (positive) correlation between MFD and market capitalization (idiosyncratic volatility), suggesting that MFD may capture information uncertainty in the crypto market, consistent with the arguments in Zhang (2006) and Bali et al. (2026).

<Insert Table 2 Here>

III. MFD and the Cross-Section of Crypto Returns

In this section, to investigate the return predictability of investor belief disagreement in the cryptocurrency market, we employ the univariate and bivariate portfolio analyses, and Fama-MacBeth (1973) regression analysis using MFD as the proxy for investor belief disagreement.

A. Univariate Portfolio Analysis

We begin by employing the univariate portfolio analysis to investigate the cross-sectional relationship between MFD and future cryptocurrency returns. For each week from January 2019 to June 2024, all cryptocurrencies are sorted into ten decile portfolios based on their MFD, where portfolio High contains cryptocurrencies with the highest MFD and portfolio Low contains cryptocurrencies with the lowest MFD. Then, we calculate (both equal-weighted and value-weighted) the time-series average of one-week-ahead excess return of each decile.

Besides, we compute two types of risk-adjusted returns using the CCAPM model with the cryptocurrency market factor (CMKT),¹¹ and Liu et al. (2022) three-factor model with CMKT, size (CSMB)¹², and momentum (CMOM)¹³. This allows us to assess whether the return spread between High-MFD and Low-MFD portfolios could be explained by standard risk factors.

Table 3 presents the results for univariate portfolio analysis. Specifically, for equal-weighted portfolios, excess returns and alphas decrease monotonically across MFD deciles. Specifically, the excess return decreases from 1.20% (t-value is 1.74) to -0.03% (t-value

¹¹ CMKT is the cryptocurrency excess market returns, defined as the difference between cryptocurrency market return and the risk-free rate. In this study, we measure the risk-free rate as the one-month Treasury bill rate.

¹² CSMB is the cryptocurrency size factor. For each week, cryptocurrencies are sorted into terciles based on market capitalization, and we form value-weighted portfolios for the bottom 30% (small-cap), the middle 40% (middle-cap), and top 30% (large-cap) groups. The CSMB is defined as the return spread between the small-cap and large-cap portfolios.

¹³ CMOM is the cryptocurrency momentum factor. For each week, cryptocurrencies are sorted into terciles based on three-week momentum, and we form value-weighted portfolios for the bottom 30% (small-mom), middle 40% (middle-mom), and top 30% (large-mom) groups. The CMOM is defined as the return spread between the small-mom and large-mom portfolios.

is -0.04) when moving from the portfolio High to the portfolio Low. The long-short portfolio that short-sells cryptocurrencies in the lowest MFD decile and takes a long position in cryptocurrencies in the highest MFD decile earns an average return of 1.22% per week, and statistically significant at the 1% level with a t-value of 2.60. Notably, the magnitude and statistical significance of the return spreads on MFD-sorted portfolios does not change after controlling for well-accepted risk factors. The CCAPM and three-factor alphas for the long-short portfolio is 1.09% with a t-values of 2.54 and 1.06% with a t-value of 2.53.

A similar MFD-return pattern holds for the value-weighted portfolios. More interestingly, we find that both return and alpha spreads on the value-weighted MFD-sorted portfolios is larger in absolute magnitude and more statistically significant than that on the equal-weighted MFD-sorted portfolios. In addition, the alphas of portfolio High are significantly negative and large in absolute magnitude across all factor models, indicating that the negative MFD-return relation is primarily concentrated on the short leg of the arbitrage portfolio. That is, cryptocurrencies with high disagreement appear to be overpriced relative to those with lower disagreement.

We note that the negative MFD-return relation is stronger based on value-weighted returns than equal-weighted returns. The cryptocurrency data quality and price/return outliers are two main concerns. While our sample selection criteria could mitigate the potential data errors and extreme outliers to some extent, concerns about the potential

dataset errors and extreme outliers may still exist. After addressing these extreme return outliers, the negative MFD-return relation is similar in terms of equal-weighted and value-weighted returns. We discuss this issue in detail in the Internet Appendix A.

Overall, these results are consistent with existing literature documenting a negative association between investor disagreement and future returns (e.g., Miller, 1977; Hong and Stein, 2007; Berkman, Dimitrov, Jain, Koch and Tice, 2009; Yu, 2011; Bali et al., 2026; Garfinkel et al., 2025).

<Insert Table 3 Here>

Next, we examine the cross-sectional persistence of MFD ranking across time. Following Atilgan et al. (2020), we estimate the average transition probabilities of a cryptocurrency remaining in or moving across MFD deciles over 12-week horizon. If MFD evolved randomly, the transition probabilities would be close to 10% for each decile.

Table 4 shows a strong persistence. For instance, 27.97% of cryptocurrencies in the lowest MFD decile (decile 1) remain in the same decile after 12 weeks, while 28.70% of those in the highest MFD decile (decile 10) remain in the highest MFD decile. Diagonal transition probabilities across other deciles range from 12.79% to 17.50%, all substantially higher than the random benchmark. Moreover, the probability of transitioning from the lowest MFD decile to the highest MFD decile or vice versa is notably lower than 10%, reinforcing the notion that MFD is a persistent characteristic in the cross-section of cryptocurrencies.

<Insert Table 4 Here>

To further investigate the persistence of MFD-based return predictability, we examine the long-term performance of MFD-based portfolios by calculating the weekly excess returns and three-factor alphas from 2 to 12 weeks after portfolio formation. As Table 5 shows, for equal-weighted portfolios, the return and three-factor alpha spreads on MFD-sorted portfolios are statistically insignificant beyond the first week. Consistent with the main results, we find a more persistent MFD-return relation in terms of value-weighted portfolio returns¹⁴. The significantly predictive power of MFD persists up to 5 weeks after the portfolio formation, indicating that large cryptocurrencies exhibit more persistent mispricing related to disagreement.

<Insert Table 5 Here>

In summary, the univariate portfolio analysis documents that the negative cross-sectional association between MFD and future cryptocurrency returns is economically meaningful and robust in both short term and over intermediate horizons, especially for value-weighted portfolios. These results underscore the relevance of investor disagreement as a key source of return predictability in cryptocurrency market.

B. Portfolio Characteristics

To address the possibility that the negative MFD-return relation could be driven by

¹⁴ The long-term performance of MFD is also affected by the potential data errors and extreme outliers. We discuss it in detail in the Internet Appendix A.

other related characteristics, we control some well-known characteristics that may be useful in explaining the cross-sectional cryptocurrency returns. Specifically, Liu and Tsyvinski (2021) and Liu et al. (2022) emphasize that market return, size, and momentum are the primary driving factors for expected cryptocurrency returns. Additionally, Garfinkel et al. (2025) document a negative association between abnormal trading volume and future cryptocurrency returns.

We also include several other prominent return predictors from the equity literature, despite evidence that their predictive power is generally weaker in the cryptocurrency market (Liu et al., 2022). These variables include volatility, idiosyncratic volatility, the maximum daily returns, and liquidity.

Before proceeding with the bivariate portfolio analysis, we investigate how these crypto-specific characteristics correlate with investor disagreement, as proxied by MFD. Specifically, we sort cryptocurrencies by MFD into ten decile portfolios each week, and compute the time-series means of the cross-sectional averages of various crypto-specific characteristic within each decile portfolio, as well as the difference between High-MFD and Low-MFD portfolios. As Table 6 shows, mean MFD increases substantially across the deciles, from 0.06 in portfolio Low to 0.31 in portfolio High, consistent with Detzel, Liu, Strauss, Zhou and Zhu (2021) that cryptocurrencies are difficult to value due to the extreme heterogeneity and information frictions.

We find that, consistent with Kumar (2009) and Bali et al. (2011), cryptocurrencies

with higher MFD have smaller market capitalization, lower liquidity, higher idiosyncratic volatility, and greater lottery-like features. These features have been interpreted as signals of elevated information uncertainty. In addition, we observe that cryptocurrencies with higher MFD exhibit higher turnover, consistent with Barinov (2014), who argues that turnover is positively related to uncertainty.

Moreover, High-MFD cryptocurrencies tend to exhibit abnormally low trading volume, stronger momentum, higher market beta, and higher overall return volatility. While no statistically significant difference in co-skewness is observed, it is retained as the control variables in subsequent analyses due to its theoretical relevance in capturing an asset's sensitivity to market-wide shocks and its potential influence on the pricing of investor disagreement.

<Insert Table 6 Here>

C. Bivariate Portfolio Analysis

Next, we examine robustness of the negative MFD-return relation by controlling for a broad set of well-known return predictors. These return predictors include momentum (*MOM*), market capitalization (*Size*), abnormal trading volume (*Abvol*), market (*Beta*), idiosyncratic volatility (*Ivol*), return volatility (*Volt*), the demand for lottery-like crypto assets (*MAX*), Co-skewness (*Coskew*), illiquidity (*ILLIQ*), and turnover (*Turnover*).

To implement the bivariate portfolio analysis, we conduct 5×10 dependent double sorts based on various crypto-specific characteristics and MFD. For each week, cryptocurrencies are sorted into five quintiles based on the ascending order of a given crypto-specific characteristic. Subsequently, we sort cryptocurrency within each crypto-specific characteristic quintile into ten decile portfolios based on MFD. This process generates 50 conditionally double-sorted portfolios for each crypto-specific characteristic. Due to space constraints, we do not report results for all 50 portfolios individually. Instead, we only report ten combined portfolios in Table 7, where portfolio Low (High) consists of cryptocurrencies with the lowest (highest) MFD in each crypto-specific characteristic quintile.

As Table 7 shows, the negative MFD-return relation remains economically meaningful and statistically significant after controlling for a broad set of well-established return predictors. For example, after controlling for momentum, we observe a return spreads of MFD-sorted portfolios is -1.01% (-1.58%) per week, with corresponding t-value of -2.30 (-2.71). The associated three-factor alpha is -0.82% (-1.21%) per week, with a t-value of -2.08 (-2.26).

Moreover, when we control for other crypto-specific characteristics, the return spreads of MFD-sorted portfolios range from -1.45% (t-value is -3.35) to -0.74% (t-value is -1.97) per week for equal-weighted portfolios, and from -2.09% (t-value is -3.03) to -1.31% (t-value is -2.09) per week for value-weighted portfolios. The three-factor alpha

spreads on MFD-sorted portfolios remain significantly negative regardless of the crypto-specific characteristic, with the exception of the idiosyncratic volatility and return volatility in the value-weighted portfolios. These findings provide strong evidence that the MFD-based return predictability is not subsumed by other cross-sectional return predictors.

<Insert Table 7 Here>

D. Fama-MacBeth Regressions

While portfolio analysis is a widely used non-parametric approach that effectively reduces noise and highlights broad patterns, it omits large amount of cross-sectional information and cannot control for multiple factors simultaneously. To address this concern, we use the Fama and MacBeth (1973) regressions to test the MFD-based return predictability. This approach allows for a more precise estimation of the partial effect of MFD while accounting for a broad set of return predictors.

Each week, we run cross-sectional regressions (i.e., across cryptocurrencies) of one-week-ahead excess returns on MFD as well as a broad set of control variables. As discussed above, we control *MOM*, *Size*, *Abvol*, *Beta*, *Ivol*, *Volt*, *MAX*, *Coskew*, *ILLIQ*, and *Turnover*. We follow Newey and West (1987) and correct the standard errors of the average slope coefficients for heteroskedasticity and autocorrelation, and report the time-series average of slope coefficients in Table 8.

As Table 8 shows, the relationship between MFD and future cryptocurrency returns

remains significantly negative across all specifications. The univariate regression in Column (1) shows that the time-series average slope coefficient of MFD is -7.80 and statistically significant at the 1% level with a t-value of -3.54 , indicating a strong inverse relationship. In columns (2-11), we sequentially include each control variable in the cross-sectional regression. While the magnitude of the MFD coefficient varies slightly, ranging from -8.70 to -5.61 , all coefficients remain significant, with t-values exceeding 2.58. Other cryptocurrency characteristics do not materially attenuate the predictive power of MFD. Overall, these results provide strong evidence that MFD contains incremental, value-relevant information about future cryptocurrency returns.

<Insert Table 8 Here>

IV. Economic Mechanisms

In this section, we further explore the underlying economic mechanisms that may drive this relation, shedding light on whether the MFD-return relation reflects mispricing or risk compensation in the crypto market.

A. The Role of Past Returns

The MFD approach allows us to identify the most important factors driving the disagreement among investors. The MFD as a flexible belief-simulation framework can accommodate various simple and sophisticated belief formation rules. Because retail

investors dominate in the crypto market (Kogan et al., 2024) and they tend to extrapolate past returns (e.g., Da et al., 2021; Giglio et al., 2021; Lin et al., 2025), we examine whether a return-based MFD that relies more on past-return variables has a strong return predictability.

Figure 2 plots the average feature importance across the 100 simulated investors obtained from the random forest forecasts used to construct MFD. Characteristics related to past returns and price dynamics emerge as the most influential predictors, broadly consistent with extrapolative behavior. Meanwhile, many non-return variables also contribute to forecast variation.

To further assess the role of return extrapolation in generating belief dispersion in the cryptocurrency market, we implement three belief-formation regimes within the MFD framework that differ in investors' degree of extrapolative beliefs. First, we model expectations as being formed purely through extrapolation from past returns. That is, we construct a return-based MFD that relies only on 9 past return variables. For comparison, we construct a non-return-based MFD that relies on other variables such as size, age, liquidity, etc. We also construct a return-plus-based MFD that 9 key past return variables and 15 non-return variables.

Table 9 reports the results. Our results show that the return-based MFD has a stronger return predictability than the non-return-based MFD. The excess return spread for the return-based MFD is -1.98% with a t -value of -3.36 , while the return spread for the non-

return based MFD is only -1.09% with a t-value of -1.81 , although the difference is insignificant. Moreover, a return-plus-based MFD that adds additional non-return information does not have a stronger return predictability than the return-based MFD, suggesting the incremental information of non-return variables is limited if we consider key past return variables in constructing MFD. We also confirm the similar results holds on 13 past return variables (Internet Appendix B). Overall, these results suggest that the MFD is primarily driven by past-return information, a pattern that is broadly consistent with extrapolative behavior documented in the cryptocurrency literature. We emphasize that our evidence should not be interpreted as directly identifying extrapolative beliefs. These findings also support the interpretation of MFD as a flexible belief-simulation framework that can accommodate behavioral heuristics such as extrapolation by varying the relative importance assigned to different information sources.

<Insert Figure 2 Here>

<Insert Table 9 Here>

B. MFD and Limits to Arbitrage

In this section, we investigate whether the MFD-based return predictability is driven by limits to arbitrage. Motivated by Miller (1977)'s hypothesis, we expect that return predictability of MFD should be more pronounced for cryptocurrencies with greater arbitrage frictions.

Shleifer and Vishny (1997) argue that arbitrage activity is risky and costly. Thus, mispricing may persist when the costs of arbitrage outweigh the potential gains. Consistent with Zhang (2006) and Lam and Wei (2011), we measure limits to arbitrage with three fundamental dimensions: (1) arbitrage risk, measured by idiosyncratic volatility; (2) information uncertainty, measured by cryptocurrency age and size; (3) transaction costs, measured by Amihud (2002)'s illiquidity. Cryptocurrencies with higher idiosyncratic volatility, smaller market capitalization, shorter trading history, and lower liquidity are more likely to face pronounced limits to arbitrage.

We first examine the return predictability of MFD conditional on different proxies for limit-to-arbitrage using bivariate portfolio analysis. Each week, cryptocurrencies are sorted into three terciles based on four limit-to-arbitrage proxies, idiosyncratic volatility, age, size, and illiquidity, respectively. Then, within each limit-to-arbitrage tercile, cryptocurrencies are further sorted into five quintile portfolios based on MFD.

As shown in Panel C-F of Table 10, the negative relation between MFD and future returns is significantly stronger among cryptocurrencies that are younger and exhibit higher return volatility. Moreover, the negative MFD-return relation is also more pronounced among illiquidity cryptocurrencies. These findings indicate that arbitrage frictions amplify the return predictability of MFD.

An exception arises for size, the negative MFD-return relation is weak and statistically insignificant among smallest cryptocurrencies. One plausible explanation is that a subset

of “small and high-MFD” cryptocurrencies experience extreme positive returns, which may obscure the underlying relation¹⁵. To address the influence of extreme outliers, we winsorize weekly returns at the 1st and 99th percentiles and repeat the bivariate portfolio analysis (see Internet Appendix C).

Consistent with expectations, after winsorization, the negative MFD-return relation is strongest among youngest, most volatile and illiquid cryptocurrencies. More importantly, the negative MFD-return relation among smallest cryptocurrencies also becomes stronger and statistically significant. Taken together, these findings are consistent with the predictions of Miller (1977).

To provide a comprehensive measure for limits-to-arbitrage, we construct a composite index using these four aforementioned proxies. Following Atilgan et al. (2020), each week, we rank cryptocurrencies in ascending order according to their idiosyncratic volatility and illiquidity, and in descending order according to their size and age. Then, cryptocurrencies are given a score of their decile rank for each limits-to-arbitrage proxy. The crypto-specific arbitrage index (ARB) is calculated as the sum of these four scores, ranging from 4 to 40, where higher values indicate greater arbitrage costs for a cryptocurrency.

We first calculate the time-series average of the cross-sectional ARB index for cryptocurrencies in MFD quintile portfolios and reported in Panel A of Table 10. We observe that the average ARB index increases sharply across MFD deciles. The difference

¹⁵ We discuss this issue in Internet Appendix A.

in ARB index between High-MFD and Low-MFD portfolios is 3.70 with a t-value of 21.78, indicating that high-MFD cryptocurrencies are more difficult to arbitrage.

Next, to explore the interaction between arbitrage costs and MFD-based return predictability, we conduct dependent double sorts on ARB and MFD. Specifically, we sort cryptocurrencies into three tercile portfolios based on their ARB index each week. Subsequently, cryptocurrencies are sorted into five MFD-based quintiles within each ARB tercile. We calculate return and three-factor alpha spreads between High-MFD and Low-MFD portfolios within each ARB tercile.

As Panel B of Table 10 shows, the return and three-factor alpha spreads on value-weighted MFD-based portfolios increase monotonically in absolute magnitude moving from low-ARB to high-ARB terciles. Specifically, the return and three-factor alpha spreads on MFD-sorted portfolios are -3.74% per week with t-value of -3.05 and -3.29% with a t-value of -3.12 for cryptocurrencies with high arbitrage cost, i.e., in the high ARB tercile. In contrast, these spreads are smaller and often insignificant in the low ARB tercile. More importantly, the difference in the return (three-factor alpha) spread between high ARB and low ARB terciles is -3.03% (-3.11%) with a t-value of -2.41 (-2.83). This result suggests that the negative cross-sectional MFD-return relation is more pronounced for cryptocurrencies with high arbitrage costs, consistent with the role of short-sale constraints in Miller (1977).

<Insert Table 10 Here>

C. MFD and Investor Sentiment

Lee and So (2015) conceptually argue that investor sentiment could better explain the origin of mispricing and limits to arbitrage could better explain the persistence of mispricing. Stambaugh, Yu, and Yuan (2012) show that stock market anomalies are more pronounced following high sentiment. Cen, Lu, and Yang (2013) examine the breadth–return relation by jointly considering sentiment and disagreement. In particular, Kim, Ryu, and Seo (2014) show that the negative relationship between investor disagreement and future returns is stronger during high sentiment in the stock market. Di Francesco and Hommes (2025) examine the sentiment-driven speculation in the cryptocurrency market in a framework of heterogeneous beliefs. In this section, we test how the predictability of MFD varies conditional on different investor sentiment.

We use the *CMC Crypto Fear and Greed Index* from CoinMarketCap as the measure of weekly investor sentiment in the cryptocurrency market.¹⁶ Following standard practice, we classify each week as either a high-sentiment or low-sentiment period based on whether the sentiment index is above or below the annual median. After that, we repeat the univariate MFD portfolio analysis separately for high-sentiment and low-sentiment periods.

As Table 11 shows, the negative MFD–return relation is more pronounced following

¹⁶ The CMC Fear and Greed Index, developed by coinmarketcap.com, serves as a widely recognized analytical tool for assessing prevailing sentiment in the cryptocurrency market. This index ranges from 1 to 100, where lower scores denote extreme fear and higher scores indicate extreme greed. The dynamics of the sentiment level help us capture the emotion state of market participants. Notably, this index also offers potential insights into whether the cryptocurrency market may be overvalued (extreme greed) or undervalued (extreme fear).

high-sentiment periods. The return spread on the value-weighted (equal-weighted) MFD-sorted portfolios is -2.87% (-2.18%) per week following high sentiment periods, and statistically significant at the 1% level with a t-value of -3.44 (-2.92). Moreover, the three-factor alpha spread is statistically and economically significant.

In contrast, the return and three-factor alpha spreads on MFD-sorted portfolios are statistically insignificant following low sentiment periods across most specifications. Notably, the return (three-factor alpha) difference between high- and low-sentiment periods are statistically significant on the equal-weighted portfolios. The return spread is economically significant based on value-weighted returns. These findings suggest that the overpricing of high MFD cryptocurrencies is concentrated during high-sentiment periods and that the cryptocurrency market exhibits behavioral dynamics similar to those documented in the stock market.

<Insert Table 11 Here>

D. Mispricing versus Risk

The negative cross-sectional MFD-return relation challenges the interpretation of disagreement as a proxy for uncertainty or risk (e.g., Merton and Others, 1987; Johnson, 2004). According to Miller (1977), when investor disagreement is high, short-sale constraints impede pessimistic views to be fully incorporated into asset prices, leading to overpricing and subsequently lower future returns. If the MFD-based return predictability

is indeed driven by mispricing, then the MFD-return relation should be more pronounced for overpriced cryptocurrencies.

To test this mispricing argument, we construct a mispricing index (MISP) by adapting the approach in Stambaugh, Yu and Yuan (2015). Given the relative immaturity of the cryptocurrency market and the limited number of established anomalies compared to traditional asset classes, we base our mispricing measure on two well-documented return predictors in the cryptocurrency literature: size and momentum (Liu et al., 2022). Each week, cryptocurrencies are independently ranked according to these two characteristics, with higher ranks indicating higher future returns, and thus, greater underpricing. The mispricing index (MISP) for each cryptocurrency is computed as the arithmetic average of its two ranks, where a higher MISP value indicates greater overpricing.

We first report the time-series average of cross-sectional MISP score for cryptocurrencies in MFD quintile portfolios. Panel A of Table 12 shows that cryptocurrencies with high MFD indeed have higher average MISP score than cryptocurrencies with low MFD. The high-minus-low difference in average MISP score is 0.34, and statistically significant at the 1% level. This provides evidence that cryptocurrencies with high disagreement is more likely to be overpriced.

To further explore whether the return predictability of MFD is concentrated in overpriced cryptocurrencies, we perform bivariate portfolio sorts. Specifically, we sort cryptocurrencies into three terciles based on MISP score each week. Subsequently,

cryptocurrencies are sorted into five quintiles based on MFD within each MISP decile.

As shown in Panel B of Table 12, the return and three-factor alpha spreads on value-weighted MFD-sorted portfolios are significantly negative and large in the absolute magnitude for overvalued cryptocurrencies, compared to the return and three-factor alpha spreads on value-weighted MFD-sorted portfolios for undervalued cryptocurrencies. In the high-MISP tercile (i.e., overvalued cryptocurrencies), the return spread MFD-sorted portfolios is -2.46% with a t-value of -3.16 , and the corresponding three-factor alpha spread on MFD-sorted portfolios is -1.87% with a t-value of -2.75 per week. However, in the low-MISP tercile (i.e., undervalued cryptocurrencies), the return and three-factor alpha spreads are insignificant. Notably, the difference in return spreads is -1.51% with t-value of -1.84 , indicating that the difference of the return spreads of the overvalued cryptocurrencies (High-MISP tercile) vs. undervalued cryptocurrencies (Low-MISP tercile) also generate difference in MFD premium.

Taken together, these results demonstrate that the negative MFD-return relationship is concentrated for the most overpriced cryptocurrencies, providing evidence consistent with a mispricing interpretation.

<Insert Table 12 Here>

V. Additional Analyses

In this section, we provide additional evidence that MFD captures investor disagreement

and have robust predictive power on cryptocurrency future returns. First, we compare MFD to turnover-based disagreement proxy in return predictability. Then, we examine the predictive power of MFD for trading volume and volatility. Finally, we conduct several robustness tests, including transaction costs, alternative construction of MFD, extensive sample periods, and alternative sample selection criteria.

A. Comparing MFD to Turnover-Based Disagreement

Existing literature employs a variety of proxies to measure investor disagreement. Trading turnover has been widely used as a disagreement proxy in both equity and cryptocurrency markets. For example, Boehme, Danielsen and Sorescu (2006) use turnover directly, while Garfinkel (2009) construct changes in market-adjusted turnover to isolate disagreement-driven trading from liquidity effects and market-wide volume trends. More recently, Garfinkel et al. (2025) employ abnormal trading volume as a proxy for investor disagreement in the cryptocurrency market.

In this section, we compare the return predictability of turnover-based disagreement measures with that of MFD. Following Garfinkel (2009), we construct the change in turnover (ΔTO) as an alternative proxy for investor belief disagreement.¹⁷ We first examine the correlation between ΔTO and MFD using portfolio sorts. Panel A of Table 13 shows a monotonic increase in ΔTO across MFD quintiles, with a statistically significant spread of 0.206 between the highest and lowest quintiles, indicating a strong positive

¹⁷ The detail construction information presented in Appendix B.

correlation between ΔTO and MFD.

To further assess the relative strength of the cross-sectional return predictability of MFD and ΔTO , we conduct a bivariate portfolio analysis. Each week, cryptocurrencies are first sorted into three terciles based on ΔTO . Within each ΔTO tercile, cryptocurrencies are then sorted into 5 quintiles based on MFD. Panel B of Table 13 reports the results. The value-weighted return and three-factor alpha spreads of MFD-sorted portfolios are statistically significant across all ΔTO terciles, with the exception of the three-factor alpha in the middle ΔTO tercile. In the highest ΔTO tercile, the return and three-factor alpha spreads are -2.11% (with a t-value of -2.86) and -1.65% (with a t-value of -2.75), respectively. However, the impact of belief disagreement is not significantly amplified when turnover-based disagreement is more severe. Although the spreads are larger in absolute magnitude in the highest ΔTO tercile than in the lowest ΔTO tercile, the differences are not statistically significant.

We next revisit the predictive power of ΔTO at different frequencies. Garfinkel et al. (2025) document that trading-based disagreement predicts daily cryptocurrency returns. Consistent with prior findings, the average daily return and alpha spreads between high- ΔTO and low- ΔTO decile portfolios are significantly negative for both equal- and value-weighted portfolios. In contrast, the average weekly return and alpha spreads are statistically insignificant for the both weighting schemes. Thus, while ΔTO exhibits predictive power at the daily frequency, its ability to predict returns dissipates at the weekly

horizon (see Internet Appendix D).

Finally, we compare the relative performance of MFD and ΔTO using cryptocurrency-level Fama-MacBeth regressions that control for other well-known return predictors. Table 14 shows that ΔTO does not exhibit predictive power for future cryptocurrency returns. More importantly, including ΔTO does not affect the statistical or economic significance of MFD.

Taken together, these findings suggest that MFD contains incremental predictive information beyond that captured by ΔTO at a weekly frequency. Turnover-based disagreement proxies, which are derived from trading activity, capture short-lived fluctuations in disagreement that tend to resolve within few days. By contrast, MFD, constructed using a 2-year rolling window and a rich set of characteristics, appears to capture a more persistent and fundamental component of belief disagreement. Moreover, because MFD is model-based and inherently forward-looking, it may better reflect differences in expectations embedded in investor beliefs. We argue that both measures are complementary in different frequencies.

<Insert Table 13 Here>

<Insert Table 14 Here>

B. Explanatory Power for Trading Volume and Volatility

A large body of literature documents that greater belief disagreement is associated

with higher trading activity (Garfinkel, 2009; Atmaz and Basak, 2018; Cookson and Niessner, 2020; Cookson, Fos, and Niessner, 2021; and Bali et al., 2026). Atmaz and Basak (2018) develop a theoretical model in which heterogenous investors become more active as belief dispersion increase, thereby generating higher trading volume. Bali et al. (2026) provide complementary empirical evidence, showing that a machine-learning-based disagreement is an important determinant of trading volume. In addition, both theoretical and empirical studies suggest that belief disagreement is positively related to market volatility (Atmaz and Basak, 2018; Bali et al., 2026).

Building on these findings from the equity market, we examine the impact of investor belief disagreement on trading volume and volatility in the cryptocurrency market. Thus, we estimate the following panel regression models.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_1 \times MFD_{i,t} + \beta_2 \times y_{i,t-1} + \gamma \times Controls_{i,t} + \epsilon_{i,t} \quad (8)$$

where $y_{i,t}$ is the weekly turnover or historical volatility of cryptocurrency i in week t . To account for persistence in turnover and volatility, we include the first lag of our dependent variable as controls. We additionally control for short-term reversal (RET), one-week momentum (MOM), cryptocurrency capitulation ($Size$), trading volume (Vol), cryptocurrency beta ($Beta$), and cryptocurrency lottery-like features (MAX). Standard errors are double-clustered by week and crypto. Notably, the effect of investor belief disagreement is captured by the coefficient β_1 .

Table 15 reports the panel regression results based on Eq. (8). In columns (1) and (3),

the coefficients on MFD are 3.53 and 15.13 and statistically significant at the 1% level, respectively, indicating a positive and economically meaningful relation between MFD and contemporaneous turnover and historical volatility. More importantly, MFD also exhibits strong predictive power for future market activity. As shown in columns (2) and (4), the coefficient on MFD is 4.42 for one-week-ahead volatility and 14.38 for one-week-ahead turnover, both statistically significant at the 1% level.

Overall, these results are consistent with the existing disagreement literature (Atmaz and Basak, 2018; Bali et al., 2026), and suggest that belief disagreement, as captured by MFD, is strongly linked not only to future returns but also to higher-order moments of the cryptocurrency market. The evidence provides evidence consistent with the view that MFD captures economically meaningful variation in investor disagreement that manifests in both trading volume and volatility.

<Insert Table 15 Here>

C. Robustness Tests

The strong negative MFD-return relation is robust to a comprehensive set of tests, which we detail in the Internet Appendix E. In traditional asset classes, many return anomalies become unprofitable once transaction costs are taken into account (Novy-Marx and Velikow, 2016), and even the most effective mitigation techniques often preserve only a small fraction of gross returns (Patton and Weller, 2020). Accordingly, we evaluate the

net profitability of MFD-based on long-short strategy. As detailed in Internet Appendix E1, the strategy remains economically profitable after accounting for conservative estimates of transaction costs.

Additionally, our results are robust to alternative specifications of the incomplete information set and alternative machine learning architectures used to construct MFD (Internet Appendix E2). To assess the temporal stability of the results, we show that the negative MFD-return relation persists across two equally split subsamples and an extended sample period (Internet Appendix E3). Finally, the MFD premium remains robust under alternative size-based screening criteria and after excluding stablecoins from the sample. (Internet Appendix E4).

VI. Conclusion

Because of the lack of a reasonable proxy for investor disagreement in the cryptocurrency market, we follow the approach in Bali et al. (2026) to construct a crypto-level measure of investor belief disagreement for cryptocurrencies. We assume that crypto investors use their individual models and common data information to form their individual return forecasts for cryptocurrencies. We find a significantly negative relation between MFD and future cryptocurrency returns in the cross section. The negative MFD-return relation holds in various robustness tests. In particular, MFD significantly predicts future returns after controlling for other disagreement proxies such as turnover.

Moreover, we find that past return variables are among the top drivers of disagreement, suggesting that MFD is broadly consistent with disagreement associated with extrapolative behavior. In addition, the negative relation is stronger for cryptocurrencies with larger limits-to-arbitrage or more severe overpricing. The negative relation is also stronger following high crypto investor sentiment periods. Our study sheds novel light on the relation between investor disagreement and future returns from the cryptocurrency market.

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Figure 1: Time Series of MFD

The figure presents the weekly time-series plot of the median cryptocurrency-level MFD. The dashed line presents the time series of the interquartile range. The sample period is from January 2019 to June 2024.

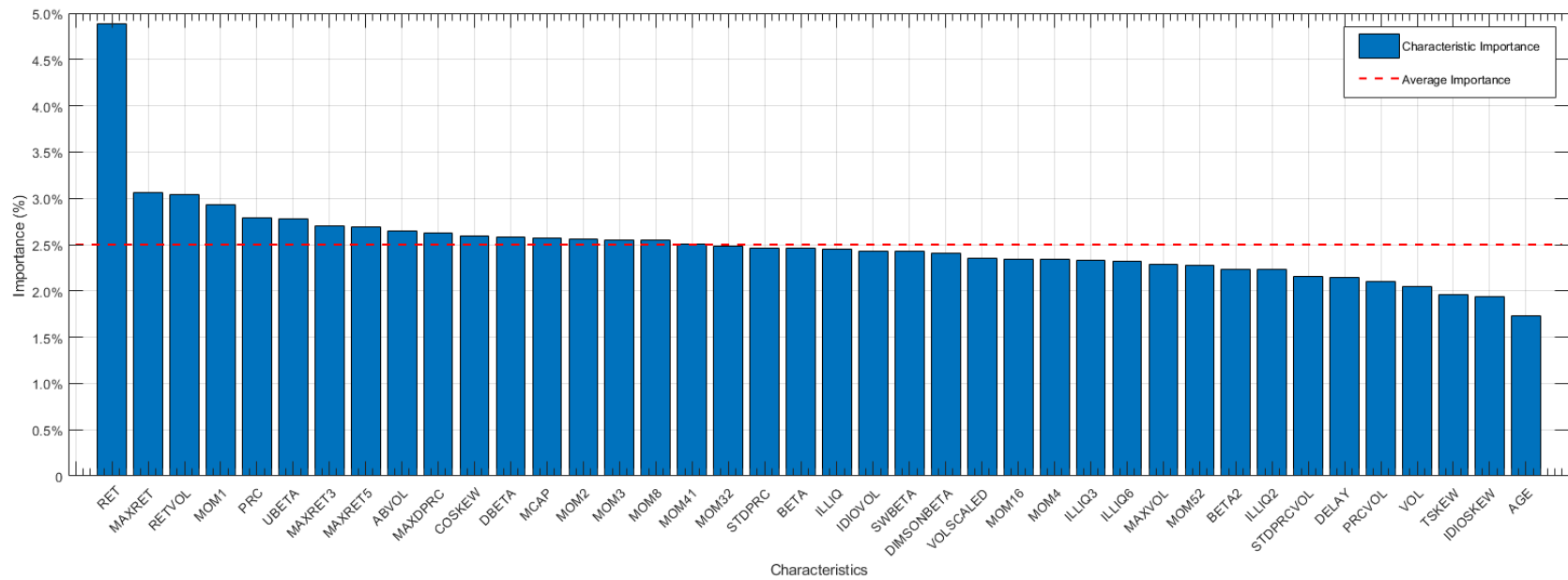


Figure 2: Feature Importance of 40 Cryptocurrency Characteristics in Random Forest Regression Prediction

This figure presents the average feature importance of 40 characteristics across 100 investors, which measures how much feature contributes to random forest regression prediction accuracy. The red dashed line represents the baseline feature importance of all 40 characteristics.

Table 1: Descriptive Statistics

This table reports the summary statistics for the cross-sectional variables. Cryptocurrencies with market capitalization below \$1 billion dollars, trading volume below \$10 thousand, price below 0.1 dollars, and trading history shorter than 52 weeks are excluded from the analysis. *Ret_1* is the one-week ahead returns in excess of the risk-free rate of individual cryptocurrency (in %). *MFD* is the machine forecast disagreement variable (scaled by 100). *Size* is the logarithm of cryptocurrency market value. *MOM* is the cryptocurrency return in the previous week (in %). *Abvol* is the natural logarithm of the average trading volume for a cryptocurrency in the portfolio formation week subtracted by the natural logarithm of the average dollar trading volume of the past 12 weeks. *Beta* is calculated by regressing the daily excess returns of each cryptocurrency on the value-weighted market excess returns during the past 90 days. *Ivol* is calculated as the standard deviation of residuals from regressing excess cryptocurrency returns on excess market returns during the past 90 days (in %). *Volt* is calculated as the standard deviation of daily return in the portfolio formation week (in %). *MAX* is the maximum daily return of the portfolio formation week (in %). *CosKew* is the coefficient of the squared excess market return term when regressing daily excess cryptocurrency returns on the daily excess market returns and the square daily excess market returns in the past 90 days. *ILLIQ* is calculated as the average absolute daily return divided by price volume during the portfolio formation week (scaled by 10^7). *Turnover* is calculated as the logarithm of the average daily volume times price scaled by market capitalization in the portfolio formation week. *ARB* is the limit-to-arbitrage index (see Section IV.B). *MISP* is the mispricing index (see Section IV.D). ΔTO is the change in turnover (an alternative proxy for investor belief disagreement, see section V.A). *AlterRF_MFD* is the MFD constructed by random forest regression using an alternative investor-specific incomplete information set, and *XGBM_MFD* is the MFD constructed by gradient boosted regression trees (see Section V.C). *PastRet_MFD* is the return-based MFD, *NonPastRet_MFD* is the Non-return-based MFD, and *PastRetPlus_MFD* is the return-plus-based MFD (see Section IV.A). All non-return variables are winsorized at the 1% and 99% levels. The mean, standard deviation (Std), minimum (Min), 10th percentile (10th), 25th percentile (25th), 50th percentile (median), 75th percentile (75th), 90th percentile (90th), and maximum (Max) are reported. The sample is from January 2019 to June 2024.

	Mean	Std	Min	10 th	25 th	Median	75 th	90 th	Max
<i>Ret_1</i>	0.931	15.638	-44.658	-11.026	-6.094	-1.130	4.921	14.091	134.016
<i>MFD</i>	0.098	0.066	0.060	0.066	0.070	0.077	0.094	0.145	0.511
<i>MOM</i>	1.590	13.496	-30.559	-10.364	-5.657	-0.664	5.615	15.637	68.641
<i>Size</i>	18.374	2.037	14.407	16.012	16.980	18.103	19.537	21.084	24.655
<i>Abvol</i>	0.017	0.486	-1.223	-0.518	-0.248	-0.017	0.229	0.583	1.772
<i>Beta</i>	0.923	0.403	-0.037	0.341	0.718	0.963	1.175	1.377	1.970
<i>Ivol</i>	5.357	3.606	0.203	2.171	3.319	4.583	6.428	9.216	23.366
<i>Volt</i>	5.249	3.508	0.133	2.084	3.287	4.532	6.289	9.019	21.992
<i>MAX</i>	8.175	7.598	0.113	2.009	3.816	6.071	9.866	16.389	46.918
<i>CosKew</i>	-2.570	3.679	-14.799	-6.841	-4.771	-2.642	-0.382	1.505	9.121
<i>ILLIQ</i>	0.355	0.644	0.000	0.001	0.013	0.068	0.328	1.022	3.713
<i>Turnover</i>	-2.572	1.409	-6.493	-4.383	-3.468	-2.519	-1.607	-0.811	0.774
<i>ARB</i>	11.493	4.366	4.000	6.183	8.477	11.493	14.718	17.695	23.525
<i>MISP</i>	7.757	2.606	3.035	4.410	5.734	7.621	9.695	11.259	13.771
<i>ΔTO</i>	-0.006	0.242	-0.906	-0.191	-0.065	-0.007	0.043	0.171	1.095
<i>AlterRF_MFD</i>	0.115	0.079	0.070	0.077	0.081	0.089	0.110	0.171	0.616
<i>XGBM_MFD</i>	0.183	0.119	0.081	0.100	0.116	0.144	0.199	0.306	0.823
<i>PastRet_MFD</i>	0.027	0.025	0.013	0.015	0.016	0.018	0.026	0.046	0.180
<i>NonPastRet_MFD</i>	0.190	0.080	0.115	0.135	0.146	0.165	0.201	0.268	0.617
<i>PastRetPlus_MFD</i>	0.082	0.045	0.053	0.059	0.061	0.067	0.081	0.122	0.338

Table 2: Cross-Sectional Correlations to MFD

This table reports the summary statistics on the correlations of various cryptocurrency characteristics with MFD. Correlation is measured using Spearman's ρ . The cryptocurrency characteristics are defined in Table 1. The mean, standard deviation (Std), 10th percentile (10th), 25th percentile (25th), 50th percentile (median), 75th percentile (75th), and 90th percentile (90th) are reported. The sample is from January 2019 to June 2024.

	Mean	Std	10 th	25 th	Median	75 th	90 th
<i>Ret_1</i>	−0.046	0.139	−0.203	−0.123	−0.055	0.032	0.130
<i>MOM</i>	0.106	0.147	−0.070	0.015	0.091	0.200	0.297
<i>Size</i>	−0.103	0.079	−0.193	−0.158	−0.111	−0.058	−0.007
<i>Abvol</i>	0.105	0.101	−0.022	0.036	0.101	0.176	0.236
<i>Beta</i>	−0.019	0.159	−0.228	−0.114	−0.010	0.097	0.178
<i>Ivol</i>	0.456	0.129	0.280	0.380	0.479	0.547	0.599
<i>Volt</i>	0.291	0.156	0.082	0.186	0.307	0.394	0.487
<i>MAX</i>	0.227	0.160	0.017	0.107	0.227	0.337	0.436
<i>CosKew</i>	−0.006	0.165	−0.188	−0.117	0.005	0.110	0.177
<i>ILLIQ</i>	0.182	0.121	0.039	0.101	0.170	0.265	0.338
<i>Turnover</i>	0.067	0.133	−0.076	−0.023	0.053	0.150	0.238

Table 3: Univariate Portfolio Sorts on MFD

This table reports excess returns and alphas on different cryptocurrency deciles sorted based on MFD and formed on a weekly basis between January 2019 and June 2024. Each week, cryptocurrencies are sorted into decile portfolios by MFD. Portfolio Low is the portfolio of cryptocurrencies with the lowest MFD, while cryptocurrencies with the highest MFD comprise portfolio High. All equal-weighted and value-weighted excess returns and alphas are a week ahead of the portfolio formation period. Excess return is the return in excess of the risk-free rate. Alpha is the intercept from a time-series regression of weekly excess returns on the factors of alternative models: CAPM and three-factor models proposed by Liu, Tsyvinski, and Wu (2022). This table also reports differences between portfolios High and Low. Newey-West (1987) adjusted t statistics are presented in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	Equal-Weighted			Value-Weighted		
	Excess Return	CAPM	3-Factor	Excess Return	CAPM	3-Factor
Low	1.196*	0.059	0.037	1.456**	0.318	0.395
	(1.74)	(0.16)	(0.10)	(2.14)	(0.91)	(1.12)
2	1.197*	0.075	0.073	0.763	-0.466	-0.514*
	(1.84)	(0.21)	(0.21)	(1.10)	(-1.47)	(-1.71)
3	1.401**	0.197	0.140	2.454***	1.072**	0.955*
	(2.13)	(0.49)	(0.38)	(2.56)	(2.08)	(1.89)
4	1.079	-0.055	-0.046	1.136	-0.152	-0.197
	(1.57)	(-0.15)	(-0.13)	(1.60)	(-0.41)	(-0.58)
5	0.847	-0.203	-0.243	1.326**	0.172	0.155
	(1.31)	(-0.57)	(-0.73)	(1.99)	(0.48)	(0.44)
6	0.917	-0.257	-0.200	1.032*	-0.0720	0.049
	(1.35)	(-0.80)	(-0.62)	(1.76)	(-0.23)	(0.16)
7	0.933	-0.158	-0.084	0.160	-0.950***	-0.643*
	(1.33)	(-0.40)	(-0.23)	(0.25)	(-2.64)	(-1.70)
8	0.920	-0.141	-0.245	0.404	-0.508	-0.487
	(1.45)	(-0.43)	(-0.78)	(0.68)	(-1.28)	(-1.26)
9	0.483	-0.484	-0.441	0.532	-0.159	-0.200
	(0.79)	(-1.31)	(-1.23)	(1.13)	(-0.48)	(-0.63)
High	-0.027	-1.028**	-1.025**	-0.565	-1.468***	-1.433***
	(-0.04)	(-2.33)	(-2.34)	(-0.90)	(-3.24)	(-3.43)
High-Low	-1.223***	-1.087**	-1.063**	-2.022***	-1.786***	-1.828***
	(-2.60)	(-2.54)	(-2.53)	(-3.29)	(-3.22)	(-3.54)

Table 4: Transition Matrix

This table reports transition probabilities for MFD at a lag of 12 weeks from January 2019 to June 2024. For each week, we sort all cryptocurrencies in our sample into deciles based on the ascending order of MFD. We then repeat this process in the week after 12 weeks. Cryptocurrencies with the lowest MFD are categorized in portfolio Low. Cryptocurrencies with the highest MFD are categorized in portfolio High. For each MFD decile, we then calculate the percentage of cryptocurrencies that fall into each of the MFD decile in the week after 12 weeks and present time-series averages of these transition probabilities in this table. Each row corresponds to a decile portfolio in a week, and each column corresponds to a decile portfolio in the week after 12 weeks.

	Low	2	3	4	5	6	7	8	9	High
Low	27.968	16.979	11.395	8.913	7.171	5.930	5.131	5.345	5.536	5.632
2	17.086	17.502	14.691	11.281	9.338	7.309	6.734	5.842	5.194	5.023
3	12.040	14.081	14.530	13.364	11.153	9.100	7.678	6.390	6.075	5.589
4	9.120	10.787	13.435	13.288	12.736	11.290	8.789	7.514	6.693	6.350
5	7.583	9.930	11.626	12.228	12.793	12.265	10.532	8.590	7.693	6.759
6	7.067	8.652	9.318	11.605	12.076	13.069	12.379	9.765	8.846	7.224
7	6.236	7.100	8.489	9.134	11.046	12.605	14.066	12.617	10.194	8.513
8	5.820	6.053	7.386	7.642	9.098	11.739	13.402	15.627	13.426	9.807
9	4.894	4.639	5.294	7.262	7.820	10.516	12.410	15.301	17.438	14.426
High	3.905	3.817	4.118	4.718	6.020	7.009	9.111	12.929	19.675	28.698

Table 5: Long-Term Portfolio Returns

This table reports the long-term predictive power of MFD. For each week $t + n$, where $n \in \{2, 3, \dots, 12\}$, individual cryptocurrencies are sorted into decile portfolios based on $week - t$ MFD. The Table reports the average weekly return and alpha (adjusted with three-factor model proposed by Liu, Tsyvinski, and Wu (2022)) of the equal-weighted and value-weighted MFD-sorted high-minus-low portfolios. Newey-West (1987) adjusted t statistics are presented in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively. The sample period is from January 2019 and June 2024.

	Equal-Weighted		Value-Weighted	
	High-Low	Alpha	High-Low	Alpha
$t + 2$	-0.084 (-0.15)	-0.235 (-0.51)	-1.555** (-2.29)	-1.184** (-2.34)
$t + 3$	0.125 (0.17)	-0.079 (-0.12)	-1.273** (-2.14)	-0.908* (-1.86)
$t + 4$	-0.301 (-0.67)	-0.130 (-0.31)	-2.057*** (-3.03)	-1.421** (-2.45)
$t + 5$	-0.252 (-0.61)	-0.095 (-0.24)	-0.980 (-1.61)	-0.728 (-1.30)
$t + 6$	1.016 (1.15)	0.950 (1.18)	0.850 (0.50)	1.128 (0.65)
$t + 7$	1.552 (1.62)	1.485* (1.79)	-0.703 (-0.87)	-0.167 (-0.23)
$t + 8$	0.589 (0.66)	0.528 (0.69)	-1.939*** (-3.41)	-1.454*** (-2.88)
$t + 9$	1.225 (1.12)	1.209 (1.25)	-0.894 (-1.58)	-0.479 (-0.93)
$t + 10$	0.490 (0.42)	0.427 (0.41)	-0.870 (-1.38)	-0.398 (-0.75)
$t + 11$	6.995 (1.13)	6.943 (1.11)	0.355 (0.40)	0.917 (1.00)
$t + 12$	8.575 (1.18)	8.657 (1.17)	0.085 (0.11)	0.103 (0.13)

Table 6: Average Cryptocurrency Characteristics of MFD-sorted Portfolio

This table reports the time-series average for MFD (Scaled by 100) and other cryptocurrency characteristics for cryptocurrency deciles sorted based on MFD and formed on a weekly basis from January 2019 to June 2024. Other cryptocurrency characteristics include *MOM* (in %), *Size*, *Abvol*, *Beta*, *Ivol* (in %), *Volt* (in %), *MAX* (in %), *CosKew*, *ILLIQ* (Scaled by 10^7), Turnover. Portfolio Low is the portfolio of cryptocurrencies with the lowest MFD, while Portfolio High is the portfolio of cryptocurrencies with the highest MFD. The last two rows report the difference between the High and Low (High-Low) and the associated Newey-West (1987) adjusted t statistics (t-stats). *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	<i>MFD</i>	<i>MOM</i>	<i>Size</i>	<i>Abvol</i>	<i>Beta</i>	<i>Ivol</i>	<i>Volt</i>	<i>MAX</i>	<i>CosKew</i>	<i>ILLIQ</i>	<i>Turnover</i>
Low	0.064	0.488	18.444	-0.044	0.891	3.986	4.549	7.487	-2.791	0.248	-2.703
2	0.068	0.590	18.649	-0.035	0.920	4.068	4.588	7.072	-2.747	0.219	-2.690
3	0.070	0.889	18.602	-0.010	0.936	4.182	4.585	6.823	-2.735	0.226	-2.638
4	0.072	0.841	18.510	-0.001	0.942	4.419	4.733	6.936	-2.710	0.250	-2.628
5	0.075	1.226	18.535	0.001	0.940	4.566	4.801	7.119	-2.542	0.282	-2.651
6	0.079	1.013	18.503	0.011	0.932	4.867	4.873	7.231	-2.515	0.291	-2.595
7	0.084	1.017	18.281	0.028	0.912	5.292	5.090	7.676	-2.499	0.365	-2.579
8	0.094	1.523	18.262	0.024	0.907	5.728	5.331	8.274	-2.308	0.384	-2.502
9	0.118	2.746	18.140	0.048	0.924	6.740	6.037	9.848	-2.206	0.446	-2.426
High	0.307	5.682	17.843	0.146	0.921	9.621	7.868	13.272	-2.639	0.633	-2.334
High-Low	0.243*** (22.17)	5.194*** (10.37)	-0.6014*** (-10.18)	0.190*** (16.47)	0.030 (1.18)	5.634*** (28.02)	3.319*** (26.63)	5.785*** (14.77)	0.152 (0.87)	0.385*** (16.11)	0.369*** (4.56)

Table 7: Bivariate Portfolio Analysis

This table reports the results from bivariate portfolio analyses based on dependent double sorts of various cryptocurrency-specific characteristics and MFD. We first form quintile portfolios every week based on a given cryptocurrency characteristic. Then, we additionally form quintile portfolios based on MFD in each cryptocurrency characteristic quintile. Portfolio Low is the combined portfolio of cryptocurrencies with the Lowest MFD in each cryptocurrency characteristic quintile, while portfolio High is the combined portfolio of cryptocurrencies with the highest MFD in each cryptocurrency characteristic quintile. The characteristics are described in Table 1. The Table show the excess return and alphas (adjusted with three-factor model proposed by Liu, Tsyvinski, and Wu (2022)) for each of the equal-weighted and value-weighted MFD-sorted high-minus-low portfolios. All Newey-West (1987) adjusted t statistics are presented in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	Equal-Weighted		Value-Weighted	
	High-Low	Alpha	High-Low	Alpha
<i>MOM</i>	-1.011** (-2.30)	-0.823** (-2.08)	-1.576*** (-2.71)	-1.210** (-2.26)
<i>Size</i>	-1.414*** (-3.31)	-1.134*** (-2.89)	-1.960*** (-2.95)	-1.407*** (-2.70)
<i>Abvol</i>	-1.254*** (-3.19)	-1.105*** (-3.06)	-2.094*** (-3.03)	-1.530*** (-2.68)
<i>Beta</i>	-1.434*** (-3.74)	-1.168*** (-3.40)	-1.666*** (-2.62)	-1.356** (-2.34)
<i>Ivol</i>	-0.743** (-1.97)	-0.568* (-1.69)	-1.475** (-2.15)	-0.716 (-1.37)
<i>Volt</i>	-0.818* (-1.90)	-0.715* (-1.77)	-1.305** (-2.09)	-0.697 (-1.26)
<i>MAX</i>	-1.016** (-2.49)	-0.875** (-2.30)	-1.411** (-2.28)	-0.925* (-1.65)
<i>CosKew</i>	-1.020*** (-2.63)	-0.858** (-2.52)	-1.991*** (-3.15)	-1.358** (-2.50)
<i>ILLIQ</i>	-1.452*** (-3.35)	-1.268*** (-3.20)	-1.706** (-2.46)	-0.976* (-1.88)
<i>Turnover</i>	-1.113*** (-2.70)	-1.007** (-2.54)	-1.391** (-2.19)	-1.338** (-2.37)

Table 8: Fama-MacBeth Cross-Sectional Regression

This table reports Fama-MacBeth cross-sectional regressions of one-week-ahead excess returns on MFD and a series of other cryptocurrency characteristics as control variables from January 2019 to June 2024. Coefficients and adjusted R2 are the time-series average from weekly Fama and MacBeth (1973) regressions. And the associated t-statistics in parentheses are adjusted with the Newey-West (1987) procedure. The control variables are described in Table 1. All non-return variables are winsorized at the 1% and 99% levels. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
<i>MFD</i>	-7.795*** (-3.54)	-8.698*** (-4.25)	-8.659*** (-4.21)	-8.621*** (-4.19)	-8.051*** (-3.98)	-6.829*** (-3.08)	-5.971*** (-2.75)	-5.606*** (-2.58)	-5.985*** (-2.76)	-6.201*** (-2.88)	-6.443*** (-2.91)
<i>MOM</i>		0.020* (1.76)	0.020* (1.78)	0.022** (1.98)	0.013 (1.23)	0.014 (1.30)	0.014 (1.20)	0.016 (1.30)	0.014 (1.17)	0.016 (1.31)	0.016 (1.36)
<i>Size</i>			-0.001 (-1.40)	-0.001 (-1.60)	-0.001 (-1.50)	-0.001** (-2.09)	-0.001** (-2.22)	-0.001** (-2.27)	-0.001** (-2.31)	-0.001* (-1.77)	-0.001* (-1.95)
<i>Abvol</i>				-0.004* (-1.91)	-0.004** (-1.98)	-0.004* (-1.90)	-0.002 (-0.76)	-0.002 (-0.93)	-0.002 (-1.04)	-0.002 (-0.93)	-0.001 (-0.68)
<i>Beta</i>					0.008 (1.22)	0.009 (1.39)	0.010* (1.69)	0.010 (1.53)	0.010 (1.64)	0.010 (1.57)	0.010 (1.56)
<i>Ivol</i>						-0.049 (-1.13)	-0.027 (-0.62)	-0.031 (-0.70)	-0.022 (-0.45)	-0.019 (-0.39)	-0.016 (-0.33)
<i>Volt</i>							-0.079** (-1.98)	-0.145 (-1.43)	-0.134 (-1.29)	-0.129 (-1.16)	-0.119 (-1.08)
<i>MAX</i>								0.030 (0.65)	0.028 (0.60)	0.020 (0.41)	0.017 (0.35)
<i>CosKew</i>									0.001 (1.05)	0.000 (0.75)	0.001 (0.83)
<i>ILLIQ</i>										0.014	0.001

										(0.72)	(0.38)
<i>Turnover</i>											−0.001
											(−0.97)
<i>Constant</i>	0.017**	0.017**	0.031**	0.034**	0.029**	0.036***	0.037***	0.036***	0.036***	0.033**	0.033**
	(2.25)	(2.38)	(2.26)	(2.38)	(2.10)	(2.67)	(2.84)	(2.80)	(2.90)	(2.39)	(2.44)
<i>Adj. R²</i>	0.021	0.038	0.047	0.055	0.087	0.097	0.109	0.122	0.130	0.140	0.146
N	81068	81068	81068	81068	81068	81068	81068	81068	81068	81068	81068

Table 9: The Role of Past Returns

This table reports the average weekly return spread between the highest and lowest decile portfolios based on return-based MFD, non-return-based MFD, and return-plus-based MFD, respectively. Return information set consists of 13 characteristics related to past returns, such as past one-week return (MOM_1), past two-week return (MOM_2), past three-week return (MOM_3), past four-week return (MOM_4), past one-to-four-week return ($MOM_{4 \rightarrow 1}$), past eight-week return (MOM_8), past sixteen-week return (MOM_{16}), past thirty-two-week return (MOM_{32}), past fifty-two-week return (MOM_{52}), maximum daily return in portfolio-formation week (MAX_1), average maximum daily return of the previous three weeks before the portfolio-formation week (MAX_3), average maximum daily return of the previous five weeks before the portfolio-formation week (MAX_5), and return on portfolio-formation week (RET). Non-return information set consists of other 27 characteristics that are not directly relative to past returns. Return-based MFD is constructed by 9 key return characteristics selected from return information set, which play more significant role in capturing extrapolative expectations in investor belief formation, including MOM_1 , MOM_2 , MOM_3 , MOM_4 , $MOM_{4 \rightarrow 1}$, MAX_1 , MAX_3 , MAX_5 , and RET . Non-return-based MFD is constructed by 9 characteristics randomly selected from non-return information set. Return-plus-based MFD is constructed by mixed characteristics, which comprise 9 key return characteristics and 15 characteristics randomly selected from non-return information set. Each week, cryptocurrencies are sorted into decile portfolios by MFD. Portfolio Low is the portfolio of cryptocurrencies with the lowest MFD, while cryptocurrencies with the highest MFD comprise portfolio High. All value-weighted excess returns and alphas are a week ahead of the portfolio formation period. Excess return is the return in excess of the risk-free rate. Alpha is the intercept from a time-series regression of weekly excess returns on the three-factor models proposed by Liu, Tsyvinski, and Wu (2022). This table also reports differences between portfolios High and Low. Besides, this table reports the return difference between return-based MFD and non-return-based MFD, and between return-plus-based MFD and non-return-based MFD. All measures are reported in percentage terms. Newey-West (1987) adjusted t statistics are presented in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	Return-based MFD		Non-return-based MFD		Return-plus-based MFD		Diff		Diff	
	(A)		(B)		(C)		(A-B)		(C-B)	
	Excess Return	Alpha	Excess Return	Alpha	Excess Return	Alpha	Excess Return	Alpha	Excess Return	Alpha
Low	0.984	−0.169	1.328*	0.098	1.615**	0.336	−0.344	−0.267	0.287	0.238
	(1.48)	(−0.55)	(1.73)	(0.23)	(2.01)	(0.74)	(−0.70)	(−0.61)	(0.75)	(0.64)
2	1.310*	0.013	1.667**	0.427	1.354*	0.064	−0.0357	−0.414	−0.313	−0.363
	(1.70)	(0.04)	(2.01)	(0.86)	(1.86)	(0.19)	(−0.61)	(−0.78)	(−0.64)	(−0.77)
3	1.088	−0.111	1.133	−0.247	1.011	−0.178	−0.045	0.136	−0.122	0.069
	(1.51)	(−0.32)	(1.46)	(−0.65)	(1.51)	(−0.64)	(−0.09)	(0.29)	(−0.28)	(0.18)
4	1.319**	0.107	1.194	−0.083	1.576**	0.353	0.125	0.190	0.382	0.436
	(1.99)	(0.39)	(1.52)	(−0.23)	(2.28)	(1.06)	(0.27)	(0.48)	(0.86)	(1.13)
5	0.499	−0.550	1.463*	0.211	1.506**	0.105	−0.965*	−0.761*	0.043	−0.107
	(0.78)	(−1.58)	(1.85)	(0.55)	(2.01)	(0.31)	(−1.85)	(−1.65)	(0.11)	(−0.27)
6	0.692	−0.314	0.845	−0.254	0.796	−0.368	−0.153	−0.061	−0.049	−0.115
	(1.01)	(−0.96)	(1.38)	(−0.84)	(1.05)	(−0.86)	(−0.34)	(−0.16)	(−0.10)	(−0.25)
7	2.390**	1.186*	0.473	−0.627*	0.491	−0.401	1.917**	1.813**	0.018	0.226
	(2.35)	(1.68)	(0.69)	(−1.79)	(0.92)	(−1.40)	(2.20)	(2.41)	(0.04)	(0.65)
8	0.760	−0.220	0.748	−0.173	0.746	−0.145	0.012	−0.046	−0.002	0.028
	(1.11)	(−0.44)	(1.27)	(−0.50)	(1.11)	(−0.37)	(0.02)	(−0.08)	(−0.00)	(0.07)
9	0.421	−0.520	0.043	−0.523	0.047	−0.604	0.378	0.002	0.004	−0.081
	(0.74)	(−1.27)	(0.08)	(−1.39)	(0.08)	(−1.64)	(0.67)	(0.00)	(0.01)	(−0.17)
High	−0.999**	−1.481***	0.236	−0.682**	−0.475	−1.016***	−1.236**	0.799	−0.711*	−0.334
	(−2.06)	(−3.56)	(0.42)	(−2.00)	(−1.05)	(−2.84)	(−2.26)	(−1.49)	(−1.75)	(−0.90)
High–Low	−1.984***	−1.312***	−1.092*	−0.780	−2.090***	−1.352**	−0.892	−0.532	−0.998*	−0.572
	(−3.36)	(−2.68)	(−1.81)	(−1.43)	(−3.06)	(−2.39)	(−1.28)	(−0.79)	(−1.85)	(−1.10)

Table 10: MFD and Limits-to-Arbitrage

This table reports results from the value-weighted bivariate portfolios based on dependent double sorts of limit-to-arbitrage measures and MFD from January 2019 to June 2024. The cryptocurrency-level limits-to-arbitrage score (ARB) is constructed using cryptocurrency size (*Size*), age (*Age*), idiosyncratic volatility (*Ivol*), and illiquidity (*ILLIQ*). A low (high) cryptocurrency size, low (high) cryptocurrency age, high (low) idiosyncratic volatility, and high (low) illiquidity indicate a high (low) arbitrage cost. To calculate it, cryptocurrencies are sorted each week into deciles according to their size, age, idiosyncratic volatility, and illiquidity. The decile ranks are attributed to each cryptocurrency increasing in their idiosyncratic volatility and illiquidity, and decreasing in their size and age. Each cryptocurrency is given the corresponding score of its decile rank for each variable. Finally, the limit-to-arbitrage score on the cryptocurrency-level is the sum of the three scores such that it ranges from 4 to 40. A low (high) ARB indicates a lower (higher) arbitrage cost. Panel A reports the time-series averages of the weekly cross-sectional average of a limit-to-arbitrage score (ARB) for MFD-sorted univariate quintile portfolios. Panel B reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts. Tercile portfolios are formed every week using ARB, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific ARB tercile. Panel C reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts of *Size* and MFD. Tercile portfolios are formed every week using *Size*, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific *Size* tercile. Panel D reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts of *Age* and MFD. Tercile portfolios are formed every week using *Age*, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific *Age* tercile. Panel E reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts of *Ivol* and MFD. Tercile portfolios are formed every week using *Ivol*, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific *Ivol* tercile. Panel F reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts of *ILLIQ* and MFD. Tercile portfolios are formed every week using *ILLIQ*, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific *ILLIQ* tercile. The last column reports the alpha adjusted by the three-factor model proposed by Liu, Tsyvinski, and Wu (2022) for each of the MFD-sorted high-minus-low portfolios. All alphas are reported in percentage terms. Newey and West's (1987) adjusted t-values are reported in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Average *ARB* in *MFD* Decile Portfolio

	Low	2	3	4	High	H-L	t-stat
ARB	10.584	10.460	10.940	12.159	14.281	3.696***	21.78

Panel B: Bivariate Portfolio Sort on *ARB*

	Low	2	3	4	High	H-L	Alpha
ARB Low	1.209* (1.82)	1.574** (2.10)	1.353** (1.99)	0.739 (1.37)	0.499 (1.05)	-0.711 (-1.29)	-0.178 (-0.44)
ARB 2	1.581** (2.17)	1.012 (1.42)	1.153 (1.64)	0.807 (1.12)	-0.105 (-0.16)	-1.686*** (-2.99)	-1.534*** (-3.00)
ARB High	2.216* (1.81)	1.174 (1.45)	0.619 (0.70)	1.052 (1.07)	-1.520** (-2.13)	-3.736*** (-3.05)	-3.291*** (-3.12)
ARB High-Low	1.007 (1.03)	-0.401 (-0.71)	-0.734 (-1.21)	0.313 (0.37)	-2.019*** (-3.17)	-3.025** (-2.41)	-3.113*** (-2.83)

Panel C: Bivariate Portfolio Sort on Size (*Size*)

	Low	2	3	4	High	H-L	Alpha
Low	1.246 (1.47)	1.375* (1.71)	0.855 (1.16)	1.119 (1.62)	0.631 (0.79)	-0.615 (-0.85)	-0.547 (-0.80)
Median	1.261* (1.75)	0.995 (1.44)	0.866 (1.19)	1.037 (1.46)	-0.249 (-0.38)	-1.510*** (-2.91)	-1.294*** (-2.63)
High	1.279* (1.85)	2.386*** (2.80)	1.160* (1.80)	0.208 (0.41)	0.334 (0.77)	-0.945 (-1.63)	-0.320 (-0.77)
High-Low	0.033 (0.05)	1.011* (1.65)	0.305 (0.67)	-0.911* (-1.80)	-0.297 (-0.41)	-0.330 (-0.34)	0.227 (0.28)

Panel D: Bivariate Portfolio Sort on Age (*Age*)

	Low	2	3	4	High	H-L	Alpha
Low	1.252 (1.64)	1.968* (1.69)	0.772 (1.02)	0.114 (0.16)	-1.117* (-1.90)	-2.369*** (-3.50)	-1.916*** (-3.34)
Median	0.794 (1.08)	0.945 (1.11)	1.466** (2.04)	0.758 (1.04)	0.569 (0.75)	-0.226 (-0.37)	-0.081 (-0.14)
High	1.188* (1.87)	1.723** (2.40)	0.752 (1.14)	0.635 (1.13)	0.465 (1.16)	-0.723 (-1.34)	-0.202 (-0.54)
High-Low	-0.064 (-0.14)	-0.244 (-0.25)	-0.020 (-0.03)	0.522 (0.88)	1.582*** (2.88)	1.646** (2.31)	1.714*** (2.72)

Panel E: Bivariate Portfolio Sort on Idiosyncratic Volatility (*Ivol*)

	Low	2	3	4	High	H-L	Alpha
Low	1.336* (1.88)	1.582** (2.14)	1.179* (1.72)	0.579 (1.01)	0.348 (0.98)	-0.988 (-1.61)	-0.333 (-0.75)
Median	0.813 (1.07)	1.191* (1.54)	2.111* (1.80)	0.880 (1.05)	0.628 (0.70)	-0.185 (-0.25)	-0.280 (-0.42)
High	0.958 (1.20)	0.426 (0.52)	-0.122 (-0.17)	0.563 (0.63)	-1.604** (-1.98)	-2.563*** (-2.79)	-2.493*** (-2.97)
High-Low	-0.378 (-0.68)	-1.156* (-1.72)	-1.301*** (-2.66)	-0.016 (-0.02)	-1.952*** (-2.64)	-1.574 (-1.50)	-2.160** (-2.45)

Panel F: Bivariate Portfolio Sort on Illiquidity (*ILLIQ*)

	Low	2	3	4	High	H-L	Alpha
Low	1.275 [*] (1.92)	2.013 ^{**} (2.33)	1.207 [*] (1.94)	0.483 (0.90)	-0.134 (-0.33)	-1.410 ^{**} (-2.56)	-0.715 [*] (-1.85)
Median	1.537 ^{**} (2.00)	1.407 [*] (1.77)	1.210 [*] (1.74)	0.745 (0.99)	-0.375 (-0.56)	-1.912 ^{***} (-2.93)	-1.529 ^{***} (-2.63)
High	1.509 [*] (1.93)	0.513 [*] (0.71)	0.427 (0.64)	1.118 (1.07)	-0.843 (-1.17)	-2.352 ^{***} (-2.76)	-1.661 ^{**} (-2.00)
High-Low	0.234 (0.47)	-1.500 ^{**} (-2.55)	-0.780 [*] (-1.84)	0.635 (0.63)	-0.709 (-0.97)	-0.942 (-1.00)	-0.946 (-0.97)

Table 11: MFD and Investor Sentiment

This table reports the one-week-ahead excess-return of portfolios with the highest MFD and lowest MFD as well as their differences when investor sentiment is high or low. Equal-weighted and value-weighted one-week-ahead excess returns of different cryptocurrency terciles are sorted based on MFD and formed on a weekly basis. This table also reports the alphas (adjusted with the three-factor model proposed by Liu, Tsyvinski, and Wu (2022)) for each of the MFD-sorted high-minus-low portfolios. We define high (low) sentiment periods as weeks whose CMC Crypto Fear and Greed Index is higher (or not higher) than the median of each year. We also report the return spread difference between the low- and high-sentiment groups. Newey-West (1987) adjusted t statistics are presented in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	Equal-Weighted		Value-Weighted	
	Low Sentiment	High Sentiment	Low Sentiment	High Sentiment
	(A)	(B)	(A)	(B)
Low	0.081	2.295**	0.328	2.569***
	(0.10)	(2.46)	(0.40)	(2.73)
2	0.334	2.048**	0.107	1.410
	(0.42)	(2.32)	(0.12)	(1.57)
3	0.565	2.226**	2.171	2.733**
	(0.66)	(2.23)	(1.53)	(2.55)
4	0.658	1.495	0.046	2.212**
	(0.80)	(1.61)	(0.05)	(1.97)
5	0.513	1.175	1.045	1.604*
	(0.66)	(1.42)	(1.18)	(1.74)
6	0.328	1.498	0.645	1.414*
	(0.41)	(1.63)	(0.80)	(1.66)
7	0.438	1.421	0.003	0.315
	(0.56)	(1.53)	(0.00)	(0.31)
8	0.640	1.195	0.225	0.581
	(0.82)	(1.46)	(0.31)	(0.68)
9	0.325	0.639	-0.071	1.126
	(-0.43)	(0.78)	(-0.12)	(1.48)
High	-0.174	0.118	-0.836	-0.298
	(-0.20)	(0.12)	(-0.88)	(-0.35)
High-Low	-0.255	-2.177***	-1.164	-2.867***
	(-0.41)	(-2.92)	(-1.43)	(-3.44)
Alpha	-0.348	-1.683**	-1.327*	-2.061**
	(-0.58)	(-2.25)	(-1.89)	(-2.51)
Excess Return Diff. (A-B)	1.923**		1.703	
Alpha Diff. (A-B)	1.335*		0.734	

Table 12: MFD and Mispricing

This table reports results from the value-weighted bivariate portfolios based on dependent double sorts of the mispricing score (MISP) and MFD from January 2019 to June 2024. Following Stambaugh et al. (2015), we construct the cryptocurrency-level mispricing score (MISP) based on cryptocurrency size and momentum anomalies (Liu, Tsyvinski, and Wu, 2022). To calculate it, cryptocurrencies are sorted each week into deciles according to their size, and momentum. The decile ranks are attributed to each cryptocurrency increasing in their momentum, and decreasing in their size. Each cryptocurrency is given the corresponding score of its decile rank for each variable. Finally, the mispricing score on the cryptocurrency-level is the average of the two scores such that it ranges from 1 to 10. Low (high) MISP indicates undervaluation (overvaluation). Panel A reports the time-series averages of the weekly cross-sectional median of a mispricing score (MISP) for MFD-sorted univariate quintile portfolios. Panel B reports excess returns of 3×5 dependent bivariate value-weighted portfolio sorts. Tercile portfolios are formed every week using MISP, and then quintile portfolios are formed based on MFD within each cryptocurrency-specific MISP tercile. The last column reports the alpha (adjusted by the three-factor model proposed by Liu, Tsyvinski, and Wu (2022)) for each of the MFD-sorted high-minus-low portfolios. Newey and West's (1987) adjusted t-values are reported in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Average MISP in MFD Decile Portfolios							
	Low	2	3	4	High	High-Low	t-stat
MISP	7.714	7.623	7.651	7.727	8.054	0.340***	5.82
Panel B: Bivariate Portfolio Sort on MISP							
	Low	2	3	4	High	H-L	Alpha
MISP Low	0.833 (1.17)	1.157 (1.52)	0.616 (0.86)	0.343 (0.55)	-0.114 (-0.19)	-0.947* (-1.82)	-0.721 (-1.53)
MISP 2	1.180 (1.64)	1.872** (2.47)	1.006 (1.58)	0.964 (1.40)	0.398 (0.62)	-0.782 (-1.24)	-0.547 (-0.99)
MISP High	2.043** (2.51)	2.322** (2.35)	1.863** (2.14)	0.666 (0.96)	-0.414 (-0.60)	-2.457*** (-3.16)	-1.866*** (-2.75)
MISP High-Low	1.210** (2.03)	1.166 (1.45)	1.246* (1.76)	0.321 (0.46)	-0.300 (-0.47)	-1.510* (-1.84)	-1.145 (-1.45)

Table 13: MFD versus Change in Turnover

Panel A reports the average change in turnover (ΔTO) of MFD-sorted univariate quintile portfolios. Low (high) ΔTO indicates a lower (higher) average change in turnover. Panel B reports 3×5 dependent bivariate value-weighted portfolio sorts. First, quintile portfolios are formed every week using ΔTO . Next decile portfolios are formed based on MFD within each cryptocurrency-specific ΔTO quintile. Newey and West (1987) adjusted t-values are reported in parentheses. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is from January 2019 to June 2024.

Panel A: Average ΔTO in MFD Decile Portfolio							
	Low	2	3	4	High	High-Low	t-stat
ΔTO	-0.059	-0.034	-0.026	0.038	0.147	0.206***	4.04
Panel B: Bivariate Portfolio Sort on ΔTO							
	Low	2	3	4	High	H-L	Alpha
ΔTO Low	1.630**	0.829	0.500	-0.067	-0.015	-1.645***	-1.045**
	(2.38)	(1.22)	(0.78)	(-0.11)	(-0.03)	(-2.70)	(-2.11)
ΔTO 2	1.049	1.701**	1.235*	1.555*	0.071	-0.978	-0.529
	(1.48)	(2.36)	(1.81)	(1.81)	(0.12)	(-1.61)	(-1.13)
ΔTO High	1.120	1.769*	1.107	0.246	-0.986	-2.106***	-1.651***
	(1.43)	(1.67)	(1.53)	(0.37)	(-1.34)	(-2.86)	(-2.75)
ΔTO High-Low	-0.510	0.941	0.607	0.313	-0.972	-0.462	-0.607
	(-0.91)	(1.10)	(1.08)	(0.51)	(-1.29)	(-0.50)	(-0.72)

Table 14: Fama-MacBeth Cross-Sectional Regressions on MFD and Turnover-Based Disagreement

This table reports Fama-MacBeth cross-sectional regression for MFD while additionally controlling for turnover-based disagreement measure. MFD, ΔTO , and other control variables in week t are matched to cryptocurrency returns in week $t + 1$. The dependent variable is the cryptocurrencies' future return in the first row. The control variables are described in Table 1. Winsorized at 1%. Newey and West (1987) adjusted t -statistics are reported in parentheses. *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively. The sample period is from January 2019 to June 2024.

	Excess Return	Excess Return	Excess Return	Excess Return
<i>MFD</i>			-7.005***	-5.345**
			(-2.99)	(-2.33)
ΔTO	-0.005	-0.004	-0.001	-0.002
	(-1.18)	(-0.97)	(-0.13)	(-0.50)
<i>MOM</i>		0.011		0.014
		(0.87)		(1.16)
<i>Size</i>		-0.001		-0.001*
		(-1.87)		(-1.81)
<i>Abvol</i>		-0.001		-0.001
		(-0.63)		(-0.39)
<i>Beta</i>		0.0104		0.009
		(1.62)		(1.45)
<i>Ivol</i>		-0.065		-0.012
		(-1.41)		(-0.23)
<i>Volt</i>		-0.121		-0.110
		(-1.06)		(-0.95)
<i>MAX</i>		0.0141		0.013
		(0.29)		(0.27)
<i>CosKew</i>		-0.000		0.000
		(-0.62)		(0.40)
<i>ILLIQ</i>		0.001		0.001
		(0.68)		(0.45)
<i>Turnover</i>		-0.001		-0.001
		(-0.83)		(-0.96)
<i>Constant</i>	0.009	0.029	0.016**	0.031**
	(1.41)	(2.00)	(2.19)	(2.19)
<i>Adj. R²</i>	0.008	0.139	0.029	0.153
<i>N</i>	79174	79174	79174	79174

Table 15: Predicting weekly Turnover and Historical Volatility

Panel A reports results for panel regression of trading volume and cryptocurrency volatility on MFD. For trading volume, we use weekly turnover as a proxy. For volatility, we use the standard deviation of daily cryptocurrencies returns in a week. We estimate the following regression $y_{i,t} = \alpha_i + \gamma_t + \beta_1 \times MFD_{i,t} + \beta_2 \times y_{i,t-1} + \gamma \times Controls_{i,t} + \epsilon_{i,t}$, where $y_{i,t}$ is either weekly turnover or the historical volatility of cryptocurrency i in week t . We also include the first lag of our dependent variable to account for persistence in volume and volatility, respectively, and add week (γ_t) and cryptocurrency (α_i) fixed effects. Additional controls ($Controls_{i,t}$) consists of the short-term reversal (Ret), momentum (MOM), cryptocurrency market capitalization (Size), trading volume (Volume), cryptocurrency beta (Beta), and lottery-like (MAX) variables. Standard errors are double-clustered by week and cryptocurrency. The dependent and independent variables are winsorized at 1% in both tails. *, **, *** denote statistical significance at the 10%, 5%, and 1% level, respectively. The sample period is from January 2019 to June 2024.

	$Volt_t$	$Volt_{t+1}$	$Turnover_t$	$Turnover_{t+1}$
MFD_t	3.532*** (15.88)	4.415*** (12.33)	15.133*** (3.09)	14.380*** (3.05)
$Volt_{t-1}$	0.037*** (9.71)			
$Volt_t$		0.340*** (11.22)		
$Turnover_{t-1}$			0.281*** (13.21)	
$Turnover_t$				0.828*** (49.89)
<i>Controls</i>	YES	YES	YES	YES
<i>Entity_fixed effect</i>	YES	YES	YES	YES
<i>Week_fixed effect</i>	YES	YES	YES	YES
<i>Obs.</i>	77,506	77,506	76,057	76,057
<i>Adj R²</i>	0.933	0.435	0.968	0.889
<i>Cluster by entity and week</i>	YES	YES	YES	YES

Appendix A: Variables for the Construction of MFD

No.	Category	predictor	Reference	Definition
1	Size	MCAP	Banz (1981)	Log last-day market capitalization in the portfolio formation week.
2	Size	PRC	Miller and Scholes (1982)	Log last-day price in the portfolio formation week.
3	Size	MAXDPRC	George and Hwang (2004)	Maximum price of the portfolio formation week.
4	Size	AGE	Barry and Brown (1984)	Log number of days listed onCoinmarketcap.com
5	Mom	R1.0	Jegadeesh and Titman (1993)	Past one-week return
6	Mom	R2.0	Jegadeesh and Titman (1993)	Past two-week return
7	Mom	R3.0	Jegadeesh and Titman (1993)	Past three-week return
8	Mom	R4.0	Jegadeesh and Titman (1993)	Past four-week return
9	Mom	R4.1	Jegadeesh and Titman (1993)	Past one-to-four-week return
10	Mom	R8.0	Jegadeesh and Titman (1996)	Past eight-week return
11	Mom	R16.0	Jegadeesh and Titman (1993)	Past 16-week return
12	Mom	R32.0	De Bondt and Thaler (1985)	Past 32-week return
13	Mom	R52.0	De Bondt and Thaler (1986)	Past 52-week return
14	Volume	VOL	Chordia, Subrahmanyam, and Anshuman (2001)	Log average daily volume in the portfolio formation week.
15	Volume	ABVOL	Chen, Jian, et al. (2022)	Log average daily volume for a cryptocurrency in the portfolio formation week is subtracted by the log average daily volume of the past 12 weeks.
16	Volume	PRCVOL	Chordia, Subrahmanyam, and Anshuman (2001)	Log average daily volume time price in the portfolio formation week.

17	Volume	VOLSCALED	Chordia, Subrahmanyam, and Anshuman (2001)	Log average daily volume times price scaled by market capitalization in the portfolio formation week.
18	Vol	BETA	Fama and MacBeth (1973)	The regression coefficient β_{CMKT}^i in $R_i - R_f = \alpha^i + \beta_{CMKT}^i(CMKT_t - R_f) + \varepsilon_i$. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.
19	Vol	BETA2	Fama and MacBeth (1973)	Beta square.
20	Vol	IDIOVOL	Ang et al. (2006)	Idiosyncratic volatility, measured as the standard deviation of the residual after estimating $R_{i,t} - R_{f,t} = \alpha^i + \beta_{CMKT}^i(CMKT_t - R_f) + \varepsilon_i$. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.
21	Vol	DBETA	Ang et al. (2006)	The market beta during the downside period: $DBETA_i = \frac{cov(R_{i,t}, CMKT_t CMKT_t < \overline{CMKT})}{var(CMKT_t CMKT_t < \overline{CMKT})}$, where $\overline{CMKT}$ is the average market returns. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.
22	Vol	UBETA	Scholes and Williams (1977)	The market beta during the upside period: $UBETA_i = \frac{cov(R_{i,t}, CMKT_t CMKT_t > \overline{CMKT})}{var(CMKT_t CMKT_t > \overline{CMKT})}$, where $\overline{CMKT}$ is the average market returns. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.

23	Vol	SWBETA	Scholes and Williams (1977)	$SWBETA_i = \frac{DBETA_i + BETA_i + UBETA_i}{1 + 2\rho}$, where ρ is the first-order serial correlation of the market portfolio's excess return.
24	Vol	DIMSONBETA	Dimson (1979)	$DIMSONBETA_i = \sum_{k=-5}^{k=5} \hat{\beta}_k^i$, where the $\hat{\beta}_k^i$ are the estimated slope coefficients from the regression model: $R_i - R_f = \alpha^i + \sum_{k=-5}^{k=5} \beta_k^i (CMKT_{t-k} - R_f) + \varepsilon_i$.
25	Vol	RETVOL	Ang et al. (2006)	The standard deviation of daily return in the portfolio formation week.
26	Vol	MAXRET	Bali Cakici and Whitelaw (2011)	Maximum daily return of the portfolio formation week.
27	Vol	MAXRET3	Bali Cakici and Whitelaw (2011)	Average maximum daily return of the previous three weeks before the portfolio formation week.
28	Vol	MAXRET5	Bali Cakici and Whitelaw (2011)	Average maximum daily return of the previous five weeks before the portfolio formation week.
29	Vol	DELAY	Hou and Moskowitz (2005)	The improvement of R2 in $R_{i,t} - R_{f,t} = \alpha^i + \beta_{CMKT}^i CMKT_t + \beta_{CMKT-1}^i CMKT_{t-1} + \beta_{CMKT-2}^i CMKT_{t-2} + \varepsilon_i$, where $CMKT_{t-1}$ and $CMKT_{t-2}$ are the lagged one- and two-day cryptocurrency market index excess returns, compared to using only current cryptocurrency market excess returns. The model is estimated using daily returns of the previous 90 days before the formation week.
30	Vol	STDPRCVOL	Chordiam Subrahmanyam and Anshuman (2001)	Log the standard deviation of price volume in the portfolio formation week.
31	Vol	STDPRC	Chordiam Subrahmanyam and Anshuman (2001)	Log the standard deviation of price in the portfolio formation week.

				The skewness of historical realized cryptocurrency returns.
32	Vol	TSKEW	Bali Engle and Murry (2016)	$TSKEW_i = \frac{\frac{1}{n} \sum_{t=1}^n (R_{i,t} - \bar{R}_i)^3}{(\frac{1}{n} \sum_{t=1}^n (R_{i,t} - \bar{R}_i)^2)^{3/2}},$ where $\bar{R}_i$ is the average periodic return of cryptocurrency i. The model is estimated using daily returns of the previous 90 days before the formation week.
33	Vol	IDIOSKEW	Boyer Mitton and Vorkink (2010)	The skewness of the residuals CAPM model regression: $IDIOSKEW_i = \frac{\frac{1}{n} \sum_{t=1}^n \varepsilon_{i,t}^3}{(\frac{1}{n} \sum_{t=1}^n \varepsilon_{i,t}^2)^{3/2}},$ where $\varepsilon_{i,t}$ are the residuals from the regression model: $R_{i,t} - R_{f,t} = \alpha^i + \beta_{CMKT}^i CMKT_t + \varepsilon_i$. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.
34	Vol	COSKEW	Harvey and Siddique (2000)	The slope coefficient on squared excess market return from a regression of excess cryptocurrency returns on the excess return of the market portfolio and the squared excess market return: $R_{i,t} - R_{f,t} = \alpha^i + \beta_{CMKT}^i CMKT_t + COSKEW_i CMKT_t^2 + \varepsilon_i$. The model is estimated using daily returns of the previous 90 days before the portfolio formation week.
35	Vol	MAXVOL	Assness, Frazzini, Gormsen, Pedersen (2020)	Maximum daily return scaled by the daily return volatility of the portfolio formation week.
36	Liquidity	ILLIQ1	Amihud (2002), Bali Engle and Murry (2016)	Average absolute daily return divided by price volume during the portfolio formation week

37	Liquidity	ILLIQ3	Amihud (2002)	Average absolute daily return divided by price volume during the past 3 weeks.
38	Liquidity	ILLIQ6	Bali Engle and Murry (2016)	Average absolute daily return divided by price volume during the past 6 weeks.
39	Liquidity	ILLIQ12	Amihud (2002)	Average absolute daily return divided by price volume during the past 12 weeks.
40	Reversal	RET	Bali Engle and Murry (2016)	The return during the portfolio formation week.

Appendix B: Change in Turnover

Following Garfinkel et al. (2006; 2009), we use the change in turnover (ΔTO) as the proxy for investor disagreement. We begin by calculating weekly turnover for each cryptocurrency each week as the average cryptocurrency daily trading volume scaled by its market capitalization. However, some cryptocurrencies may exhibit more trading in some weeks because a macroeconomic event creates significant trading in the whole market. We therefore, subtract market-wide turnover calculated the same way, but across all cryptocurrencies in the cryptocurrency market. This creates our weekly market-adjusted turnover measure.

Then, cryptocurrencies with relatively higher turnover at some weeks may reasonably be the same cryptocurrencies with relatively higher turnover overall. Thus, market-adjusted turnover may capture more than just volume attributable to investor disagreements; it can also include liquidity trading. We therefore market-adjusted turnover is subtracted by market-adjusted turnover averaged over the past 48 weeks.

$$\Delta TO = (TO_{i,t} - TO_{MKT,t}) - \frac{1}{48} \sum_{j=t-48}^{t-1} (TO_{i,t} - TO_{MKT,t}) \quad (B.1)$$